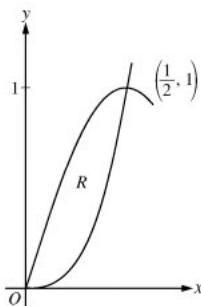


8-11-12

1.



Let R be the region in the first quadrant enclosed by the graphs of $f(x) = 8x^3$ and $g(x) = \sin(\pi x)$, as shown in the figure above.

(a) Write an equation for the line tangent to the graph of f at $x = 1/2$.

(b) Find the area of R .

(c) Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 1$.

Part A

1 point is earned for $f'(\frac{1}{2})$

1 point is earned for the answer

$$f\left(\frac{1}{2}\right) = 1$$

$$f'(x) = 24x^2, \text{ so } f'\left(\frac{1}{2}\right) = 6$$

An equation for the tangent line is $y = 1 + 6\left(x - \frac{1}{2}\right)$.

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for $f'(\frac{1}{2})$

1 point is earned for the answer

$$f\left(\frac{1}{2}\right) = 1$$

$$f'(x) = 24x^2, \text{ so } f'\left(\frac{1}{2}\right) = 6$$

An equation for the tangent line is $y = 1 + 6\left(x - \frac{1}{2}\right)$.

Part B

1 point is earned for the integrand

2 points are earned the antiderivative

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_0^{1/2} (g(x) - f(x)) dx \\
 &= \int_0^{1/2} (\sin(\pi x) - 8x^3) dx \\
 &= \left[-\frac{1}{\pi} \cos(\pi x) - 2x^4 \right]_{x=0}^{x=1/2} \\
 &= -\frac{1}{8} + \frac{1}{\pi}
 \end{aligned}$$

8-11-12



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

2 points are earned the antiderivative

1 point is earned for the answer

$$\begin{aligned}
 \text{Area} &= \int_0^{1/2} (g(x) - f(x)) dx \\
 &= \int_0^{1/2} (\sin(\pi x) - 8x^3) dx \\
 &= \left[-\frac{1}{\pi} \cos(\pi x) - 2x^4 \right]_{x=0}^{x=1/2} \\
 &= -\frac{1}{8} + \frac{1}{\pi}
 \end{aligned}$$

Part C

1 point is earned for limits and constant

2 points are earned for integrand

$$\begin{aligned}
 &\pi \int_0^{1/2} \left((1 - f(x))^2 - (1 - g(x))^2 \right) dx \\
 &= \pi \int_0^{1/2} \left((1 - 8x^3)^2 - (1 - \sin(\pi x))^2 \right) dx
 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

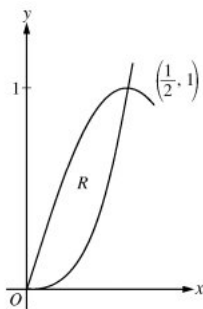
1 point is earned for limits and constant

2 points are earned for integrand

$$\begin{aligned}
 &\pi \int_0^{1/2} \left((1 - f(x))^2 - (1 - g(x))^2 \right) dx \\
 &= \pi \int_0^{1/2} \left((1 - 8x^3)^2 - (1 - \sin(\pi x))^2 \right) dx
 \end{aligned}$$

8-11-12

2.



Let R be the region in the first quadrant enclosed by the graphs of $f(x) = 8x^3$ and $g(x) = \sin(\pi x)$, as shown in the figure above.

Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 1$.

Part C

1 point is earned for limits and constant

2 points are earned for integrand

$$\begin{aligned} & \pi \int_0^{\frac{1}{2}} \left((1 - f(x))^2 - (1 - g(x))^2 \right) dx \\ &= \pi \int_0^{\frac{1}{2}} \left((1 - 8x^3)^2 - (1 - \sin(\pi x))^2 \right) dx \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

2 points are earned for integrand

$$\begin{aligned} & \pi \int_0^{\frac{1}{2}} \left((1 - f(x))^2 - (1 - g(x))^2 \right) dx \\ &= \pi \int_0^{\frac{1}{2}} \left((1 - 8x^3)^2 - (1 - \sin(\pi x))^2 \right) dx \end{aligned}$$

8-11-12

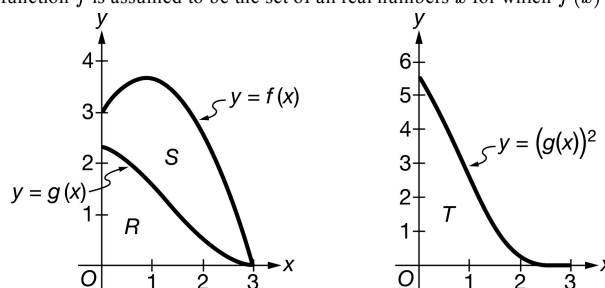
3. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



The function f is defined by $f(x) = 3(1+x)^{0.5} \cos\left(\frac{\pi x}{6}\right)$ for $0 \leq x \leq 3$. The function g is continuous and decreasing for $0 \leq x \leq 3$ with $g(3) = 0$.

The figure above on the left shows the graphs of f and g and the regions R and S . R is the region bounded by the graph of g and the x - and y -axes. Region R has area 3.24125. S is the region bounded by the y -axis and the graphs of f and g .

The figure above on the right shows the graph of $y = (g(x))^2$ and the region T . T is the region bounded by the graph of $y = (g(x))^2$ and the x - and y -axes. Region T has area 5.32021.

(a) Find the area of region S .

(b) Find the volume of the solid generated when region S is revolved about the horizontal line $y = -3$.

(c) Region S is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a rectangle whose height is 7 times the length of its base in region S . Write, but do not evaluate, an integral expression for the volume of this solid.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point can be earned with either $\int_0^3 (f(x) - g(x)) dx$ or $\int_0^3 f(x) dx$.

The second point is earned for evidence of 3.24125 being used as an area, which means it is being subtracted outside of an integrand. For example,

$\int_0^3 f(x) dx - 3.24125$ would earn the point, while $\int_0^3 f(x) - 3.24125$ would provide evidence for earning this point only in the presence of the final answer of 4.920 (or 4.919).

Note that responses in degree mode, which lead to a final answer of approximately 10.756727, do not earn the first point for which they are eligible. Thus a fully correct solution in degree mode would earn two of the three points.



0	1	2	3
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The student response accurately includes **all three** of the criteria below.

- ☐ definite integral of f
- ☐ uses area of R
- ☐ answer

Solution:

8-11-12

$$\begin{aligned}\int_0^3 (f(x) - g(x)) \, dx &= \int_0^3 f(x) \, dx - \int_0^3 g(x) \, dx \\ &= \int_0^3 f(x) \, dx - 3.24125 = 4.919585\end{aligned}$$

The area of region S is 4.920 (or 4.919).

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for an integrand of the form $(f(x) - k)^2 - (g(x) - k)^2$ where $k = -3$, 0, or 3.

The second point can be earned if f and g are reversed.

The third point is earned for replacing $\int_0^3 g(x) \, dx$ with 3.24135 and $\int_0^3 (g(x))^2 \, dx$ with 5.32021. There are multiple ways that a response can use these values correctly, so the third point can also be earned with sufficient work in the presence of the correct final value of 156.263709.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ form of integrand
- ☐ integrand
- ☐ uses areas of R and T
- ☐ limits, constant, and answer

Solution:

$$\begin{aligned}&\pi \int_0^3 ((f(x) + 3)^2 - (g(x) + 3)^2) \, dx \\ &= \pi \left(\int_0^3 (f(x) + 3)^2 \, dx - \int_0^3 ((g(x))^2 + 6g(x) + 9) \, dx \right) \\ &= \pi \left(\int_0^3 (f(x) + 3)^2 \, dx - \int_0^3 (g(x))^2 \, dx - 6 \int_0^3 g(x) \, dx - \int_0^3 9 \, dx \right) \\ &= \pi \left(\int_0^3 (f(x) + 3)^2 \, dx - 5.32021 - 6 \cdot 3.24125 - 9 \cdot 3 \right) \\ &= 156.263709\end{aligned}$$

The volume of the solid is 156.264 (or 156.263).

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned with an integrand of the form $(f(x) - g(x))^2$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ integrand
- ☐ expression

Solution:

$$\text{Volume} = \int_0^3 7(f(x) - g(x))^2 \, dx$$

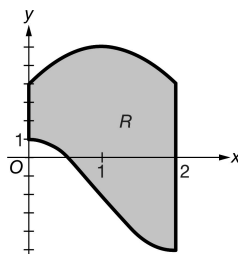
8-11-12

4. NO CALCULATOR IS ALLOWED FOR THIS QUESTION.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let R be the region enclosed by the graphs of $g(x) = -2 + 3 \cos\left(\frac{\pi}{2}x\right)$ and $h(x) = 6 - 2(x-1)^2$, the y -axis, and the vertical line $x = 2$, as shown in the figure above.

- Find the area of R .
- Region R is the base of a solid. For the solid, at each x the cross section perpendicular to the x -axis has area $A(x) = \frac{1}{x+3}$. Find the volume of the solid.
- Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 6$.

Part A

Select a point value to view scoring criteria, solutions, and/or examples to score the response.

The first point is earned for an integrand that contains a form of $h(x) - g(x)$ or $g(x) - h(x)$.

The second point is earned for the antiderivative $\frac{6}{\pi} \sin\left(\frac{\pi}{2}x\right)$.

A response is eligible for the fourth point if (1) the first point is earned, (2) at least one of the second or third points is earned, (3) the integrand contains no copy errors or arithmetic errors. Simplification is not required to earn the fourth point. The limits of integration are part of the fourth point.

If the limits of integration are $x = 1$ to $x = 2$ in all parts of the question, the response is not eligible for the fourth point in Part A but eligible for points in Part B and Part C.



0	1	2	3	4
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The student response accurately includes **all four** of the criteria below.

- ☐ integrand
- ☐ antiderivative of $3 \cos\left(\frac{\pi}{2}x\right)$
- ☐ antiderivative of remaining terms
- ☐ answer

Solution:

$$\begin{aligned}
 \int_0^2 (h(x) - g(x)) \, dx &= \int_0^2 \left((6 - 2(x-1)^2) - (-2 + 3 \cos(\frac{\pi}{2}x)) \right) \, dx \\
 &= \left[\left(6x - \frac{2}{3}(x-1)^3 \right) - \left(-2x + \frac{6}{\pi} \sin(\frac{\pi}{2}x) \right) \right]_{x=0}^{x=2} \\
 &= \left(\left(12 - \frac{2}{3} \right) - (-4 + 0) \right) - \left(\left(0 + \frac{2}{3} \right) - (0 + 0) \right) \\
 &= 12 - \frac{2}{3} + 4 - \frac{2}{3} = \frac{44}{3}
 \end{aligned}$$

The area of R is $\frac{44}{3}$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-11-12

The first point is earned for the definite integral $\int_0^2 A(x) \, dx$ or $\int_0^2 \frac{1}{x+3} \, dx$.



0	1	2
---	---	---

The student response accurately includes **both** of the criteria below.

- ☐ integral
- ☐ answer

Solution:

$$\begin{aligned}\int_0^2 A(x) \, dx &= \int_0^2 \frac{1}{x+3} \, dx \\ &= [\ln(x+3)]_{x=0}^{x=2} = \ln 5 - \ln 3\end{aligned}$$

The volume of the solid is $\ln 5 - \ln 3$.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

The first point is earned for a definite integral, multiplied by π , with 0 and 2 as the lower and upper limits of integration, respectively. The π may appear in the integrand, but it must be correct to earn the first point.

The second point is earned with an integrand that is a difference of squares with a correct or consistent axis of rotation.

The third point is earned for the correct integrand. Note that a reversal of order within the squared terms, e.g. $\pi \int_0^2 \left((g(x) - 6)^2 - (h(x) - 6)^2 \right) dx$, is equivalent to the given solution and thus earns the third point.



0	1	2	3
---	---	---	---

The student response accurately includes **all three** of the criteria below.

- ☐ limits and constant
- ☐ form of integrand
- ☐ integrand

Solution:

$$\pi \int_0^2 \left((6 - g(x))^2 - (6 - h(x))^2 \right) dx$$

8-11-12

5. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.

Let R be the region in the first quadrant enclosed by the graphs of $y = e^{(x^2)} - 1$ and $y = 4x$.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- Region R forms the base of a solid whose cross sections perpendicular to the y -axis are squares. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.

Part A

The first point requires a definite integral with a correct integrand. At most 2 out of 3 points earned for a copy error in the integrand **OR** a reversal of terms in the difference in the integrand. Subsequent points may be earned.

The second point requires correct antidifferentiation.

The third point is for limits and evaluation.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ integral
- ☐ antiderivative
- ☐ answer

Solution:

$$\text{Area} = \int_0^4 (4x - x^2) dx = \left(2x^2 - \frac{x^3}{3} \right) \bigg|_{x=0}^{x=4} = 32 - \frac{64}{3} = \frac{32}{3}$$

Part B

The first point requires a difference of squares for the integrand.

The second point requires both terms of the integrand.

At most 1 out of 2 integrand points earned if there is one error in the integrand **OR** a reversal of terms.

The third point requires the limits and π .

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ integrand of difference of squares

8-11-12

- ☐ integrand terms
- ☐ limits and π

Solution:

$$\text{Volume} = \pi \int_0^4 \left((16 - x^2)^2 - (16 - 4x)^2 \right) dx$$

Part C

The first point requires an integrand in terms of y that addresses the length of a side of the square cross section.

The second point requires the integrand to represent the cross sectional area.

At most 1 out of 2 integrand points earned if there is one error in the integrand.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ integrand in terms of y for square cross section
- ☐ integrand terms
- ☐ limits

Solution:

$$\text{Volume} = \int_0^{16} \left(\sqrt{y} - \frac{y}{4} \right)^2 dy$$

8-11-12

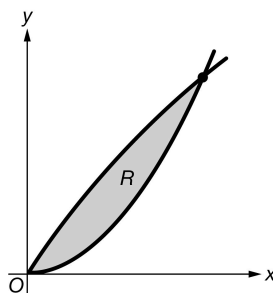
6. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let R be the region in the first quadrant enclosed by the graphs of $y = \arctan x$ and $y = x^2$, as shown in the figure above.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- Region R is the base of a solid. For the solid, each cross section perpendicular to the y -axis is a rectangle whose height is twice the length of its base in region R . Find the volume of the solid.

Part A

The first point requires a definite integral with a correct integrand.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ integrand
- ☐ limits and answer

Solution:

The graphs of $y = \arctan x$ and $y = x^2$ intersect at $(x_1, y_1) = (0, 0)$ and $(x_2, y_2) = (0.833606, 0.694899)$.

$$\text{Area} = \int_{x_1}^{x_2} (\arctan x - x^2) dx = 0.122$$

Part B

The first point requires a definite integral with a difference of squares for the integrand.

The second point requires both terms of the integrand.

At most 1 out of 2 integrand points earned if there is one error in the integrand OR a reversal of terms.

A response that imports incorrect limits from part (a) is eligible for all points.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-11-12

✓

0	1	2	3
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The student response accurately includes all three of the criteria below.

- ☐ integrand of difference of squares
- ☐ integrand terms
- ☐ limits, constant, and answer

Solution:

From part (a), $x_1 = 0$ and $x_2 = 0.833606$.

$$\text{Volume} = \pi \int_{x_1}^{x_2} \left((\arctan x)^2 - (x^2)^2 \right) dx = 0.073\pi = 0.230 \text{ (or } 0.229)$$

Part C

The first point requires an integrand in terms of y that addresses the length of either side of the rectangular cross section.

The second point requires the integrand to represent the cross sectional area.

At most 1 out of 2 integrand points earned if there is one error in the integrand.

A response is eligible for all points if the limits are consistent with the x -values from part (a).

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2	3	4
---	---	---	---	---

The student response accurately includes all four of the criteria below.

- ☐ integrand in terms of y for rectangular cross section
- ☐ integrand terms
- ☐ limits
- ☐ answer

Solution:

From part (a), $y_1 = 0$ and $y_2 = 0.694899$.

$$\text{Volume} = \int_{y_1}^{y_2} 2(\sqrt{y} - \tan y)^2 dy = 0.050 \text{ (or } 0.049)$$

8-11-12

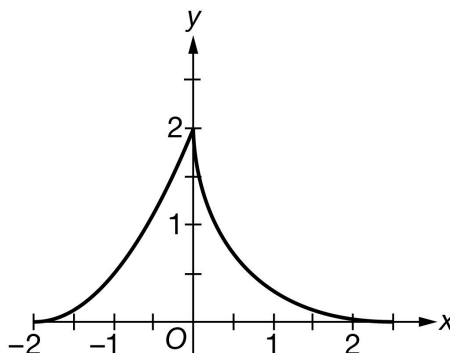
7. A GRAPHING CALCULATOR IS REQUIRED FOR THIS QUESTION.

You are permitted to use your calculator to solve an equation, find the derivative of a function at a point, or calculate the value of a definite integral. However, you must clearly indicate the setup of your question, namely the equation, function, or integral you are using. If you use other built-in features or programs, you must show the mathematical steps necessary to produce your results. Your work must be expressed in standard mathematical notation rather than calculator syntax.

Show all of your work, even though the question may not explicitly remind you to do so. Clearly label any functions, graphs, tables, or other objects that you use. Justifications require that you give mathematical reasons, and that you verify the needed conditions under which relevant theorems, properties, definitions, or tests are applied. Your work will be scored on the correctness and completeness of your methods as well as your answers. Answers without supporting work will usually not receive credit.

Unless otherwise specified, answers (numeric or algebraic) need not be simplified. If your answer is given as a decimal approximation, it should be correct to three places after the decimal point.

Unless otherwise specified, the domain of a function f is assumed to be the set of all real numbers x for which $f(x)$ is a real number.



Let f be the function defined by $f(x) = \begin{cases} \frac{1}{2}(x+2)^2 & \text{for } -2 \leq x < 0 \\ 2 - 2\sin\sqrt{x} & \text{for } 0 \leq x \leq \frac{\pi^2}{4} \end{cases}$. The graph of f is shown in the figure above. Let R be the region bounded by the graph of f and the x -axis.

- Find the area of R .
- Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an expression involving one or more integrals that gives the volume of the solid.
- The portion of the region R for $1 \leq y \leq 2$ is revolved about the x -axis to form a solid. Find the volume of the solid.

Part A

There are closed form expressions for the antiderivatives. The response may give an exact value for the area, which is $\frac{4}{3} + \frac{1}{2}(\pi^2 - 8)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2	3
---	---	---	---

The student response accurately includes all three of the criteria below.

- ☐ definite integral for $-2 \leq x < 0$
- ☐ definite integral for $0 \leq x \leq \frac{\pi^2}{4}$
- ☐ answer

Solution:

$$\text{Area} = \int_{-2}^0 \frac{1}{2}(x+2)^2 dx + \int_0^{\frac{\pi^2}{4}} (2 - 2\sin\sqrt{x}) dx = 2.268$$

Part B

At most 1 out of 2 points earned if there is one error in the integrand OR 1 error with limits of integration.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

8-11-12

✓

0	1	2
---	---	---

The student response accurately includes both of the criteria below.

- ☐ integrands
- ☐ expression for total volume

Solution:

$$\text{Volume} = \int_{-2}^0 \left(\frac{1}{2}(x+2)^2 \right)^2 dx + \int_0^{\frac{\pi^2}{4}} (2 - 2 \sin \sqrt{x})^2 dx$$

Part C

To be eligible for the fourth point, the response must have earned the first two points.

Special case: A response with the expression

$$\pi \int_{-0.585786}^0 \left(\frac{1}{2}(x+2)^2 \right)^2 dx + \pi \int_0^{0.274156} (2 - 2 \sin \sqrt{x})^2 dx = 5.688 \text{ (or } 5.687)$$

earns the third and fourth points.

There are closed form expressions for the antiderivatives. The response may give an exact values for the limits and volume. The limits are $x_1 = \sqrt{2} - 2$ and $x_2 = \frac{\pi^2}{36}$. An unsimplified expression for the volume is given by $\frac{\pi}{20} \left(2^5 - (\sqrt{2})^5 \right) + \pi (\sqrt{2} - 2) + \frac{\pi}{36} (-270 + 42\sqrt{3}\pi + 5\pi^2)$.

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

✓

0	1	2	3	4
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The student response accurately includes all four of the criteria below.

- ☐ integrand for one region
- ☐ integrand for the remaining region
- ☐ limits
- ☐ constant and answer

Solution:

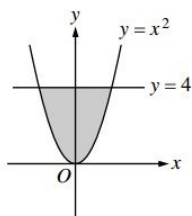
$$1 = \frac{1}{2}(x_1 + 2)^2 \Rightarrow x_1 = -0.585786$$

$$1 = 2 - 2 \sin \sqrt{x_2} \Rightarrow x_2 = 0.274156$$

$$\begin{aligned} \text{Volume} &= \pi \int_{x_1}^0 \left(\left(\frac{1}{2}(x+2)^2 \right)^2 - 1^2 \right) dx + \pi \int_0^{x_2} \left((2 - 2 \sin \sqrt{x})^2 - 1^2 \right) dx \\ &= 2.986 \text{ (or } 2.985) \end{aligned}$$

8-11-12

8.



The shaded region, R , is bounded by the graph of $y = x^2$ and the line $y = 4$, as shown in the figure above.

(a) Find the area of R .

(b) Find the volume of the solid generated by revolving R about the x -axis.

(c) There exists a number k , $k > 4$, such that when R is revolved about the line $y = k$, the resulting solid has the same volume as the solid in part (b). Write, but do not solve, an equation involving an integral expression that can be used to find the value of k .

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_{-2}^2 (4 - x^2) dx \\ &= 2 \int_0^2 (4 - x^2) dx \\ &= 2 \left[4x - \frac{x^3}{3} \right]_0^2 \\ &= \frac{32}{3} = 10.666 \text{ or } 10.667 \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_{-2}^2 (4 - x^2) dx \\ &= 2 \int_0^2 (4 - x^2) dx \\ &= 2 \left[4x - \frac{x^3}{3} \right]_0^2 \\ &= \frac{32}{3} = 10.666 \text{ or } 10.667 \end{aligned}$$

Part B

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for the answer

8-11-12

$$\begin{aligned}
 \text{Volume} &= \pi \int_{-2}^2 (4^2 - (x^2)^2) dx \\
 &= 2\pi \int_0^2 (16 - x^4) dx \\
 &= 2\pi \left[16x - \frac{x^5}{5} \right]_0^2 \\
 &= \frac{256\pi}{5} = 160.849 \text{ or } 160.850
 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for the answer

$$\begin{aligned}
 \text{Volume} &= \pi \int_{-2}^2 (4^2 - (x^2)^2) dx \\
 &= 2\pi \int_0^2 (16 - x^4) dx \\
 &= 2\pi \left[16x - \frac{x^5}{5} \right]_0^2 \\
 &= \frac{256\pi}{5} = 160.849 \text{ or } 160.850
 \end{aligned}$$

Part C

1 point is earned for limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the equation

$$\pi \int_{-2}^2 [(k - x^2)^2 - (k - 4)^2] dx = \frac{256\pi}{5}$$

✓

0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for limits and constant

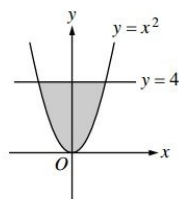
2 points are earned for integrand

<-1> each error

1 point is earned for the equation

$$\pi \int_{-2}^2 [(k - x^2)^2 - (k - 4)^2] dx = \frac{256\pi}{5}$$

8-11-12



The shaded region, R , is bounded by the graph of $y = x^2$ and the line $y = 4$, as shown in the figure above.

9. Find the volume of the solid generated by revolving R about the x -axis.

Part B

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for the answer

$$Volume = \pi \int_{-2}^2 (4^2 - (x^2)^2) dx$$

$$= 2\pi \int_0^2 (16 - x^4) dx$$

$$= 2\pi \left[16x - \frac{x^5}{5} \right]_0^2$$

$$= \frac{256\pi}{5} = 160.849 \text{ or } 160.850$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

1 point is earned for integrand

1 point is earned for the answer

$$Volume = \pi \int_{-2}^2 (4^2 - (x^2)^2) dx$$

$$= 2\pi \int_0^2 (16 - x^4) dx$$

$$= 2\pi \left[16x - \frac{x^5}{5} \right]_0^2$$

$$= \frac{256\pi}{5} = 160.849 \text{ or } 160.850$$

8-11-12

10. There exists a number k , $k > 4$, such that when R is revolved about the line $y = k$, the resulting solid has the same volume as the solid in part (b). Write, but do not solve, an equation involving an integral expression that can be used to find the value of k .

Part C

1 point is earned for limits and constant

2 points are earned for integrand

< -1 > each error

1 point is earned for the equation

$$\pi \int_{-2}^2 \left[(k - x^2)^2 - (k - 4)^2 \right] dx = \frac{256\pi}{5}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

2 points are earned for integrand

< -1 > each error

1 point is earned for the equation

$$\pi \int_{-2}^2 \left[(k - x^2)^2 - (k - 4)^2 \right] dx = \frac{256\pi}{5}$$

8-11-12

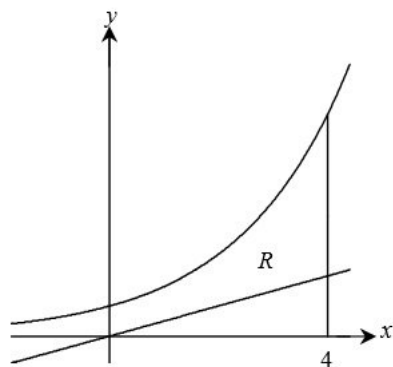
11. Let R be the region enclosed by the graphs of $y = e^x$, $y = x$ and the lines $x = 0$ and $x = 4$.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part A

Using Geometry (Area of Triangle) 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for the correct integrand



1 point is earned for the correct antidifferentiation and evaluation

0/1 if integrand is not $e^x - x$ or $x - e^x$

$$\begin{aligned}
 \text{Area} &= \int_0^4 e^x - x \, dx \\
 &= e^x - \frac{x^2}{2} \Big|_0^4 \\
 &= \left(e^4 - \frac{16}{2} \right) - (e^0 - 0) \\
 &= e^4 - 9 \\
 &\approx 45.598
 \end{aligned}$$

OR

Using y -axis 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for answering all 3 integrands correct

Note: 0/2 if any limits incorrect

$$\int_0^1 y \, dy + \int_1^4 y - \ln y \, dy + \int_4^{e^4} 4 - \ln y \, dy$$

1 point is earned for the correct answer

≈ 45.598

Note: $k \int_0^4 e^x - x \, dx, k \neq 1$

Student scores 1 - 0 on first two points and is eligible for third point

Note: Student who believes $y = ex$ meets $y = x$

$$\int_0^k e^x - x \, dx + \int_k^4 x - e^x \, dx, 0/2 \text{ Integral}$$

8-11-12

Eligible for third point whether k is a number or parameter

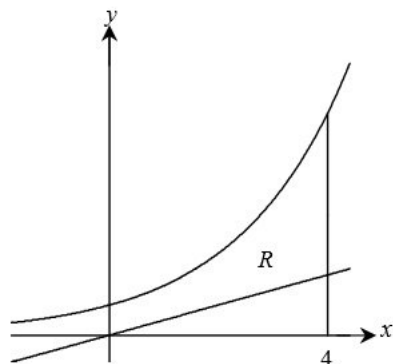
✓

0	1	2	3
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The student response earns all of the following points:

Using Geometry (Area of Triangle) 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for the correct integrand



1 point is earned for the correct antidifferentiation and evaluation

0/1 if integrand is not $ex - x$ or $x - ex$

$$\begin{aligned}
 \text{Area} &= \int_0^4 e^x - x \, dx \\
 &= e^x - \frac{x^2}{2} \Big|_0^4 \\
 &= \left(e^4 - \frac{16}{2} \right) - (e^0 - 0) \\
 &= e^4 - 9 \\
 &\approx 45.598
 \end{aligned}$$

OR

Using y-axis 3/3:

2 points are earned for the correct integral where one point is earned for the correct limits and one point is earned for answering all 3 integrands correct

Note: 0/2 if any limits incorrect

$$\int_0^1 y \, dy + \int_1^4 y - \ln y \, dy + \int_4^{e^4} 4 - \ln y \, dy$$

1 point is earned for the correct answer

≈ 45.598

Note: $k \int_0^4 e^x - x \, dx, k \neq 1$

Student scores 1 - 0 on first two points and is eligible for third point

Note: Student who believes $y = ex$ meets $y = x$

$$\int_0^k e^x - x \, dx + \int_k^4 x - e^x \, dx, 0/2 \text{ Integral}$$

Eligible for third point whether k is a number or parameter

8-11-12

Part B

Using Geometry (Volume of Cone) 4/4:

3 points are earned for the correct integral where one point is earned for π and correct limits and two points are earned for correctly using $(e^x)^2 - (x)^2$

<1> $(e^x - x)^2$

<1> $e^{x^2} - x^2$

<1> $(x)^2 - (e^x)^2$

<1> e^{x^2}

$$V_x = \pi \int_0^4 (e^x)^2 - (x)^2 dx$$

$$= \pi \int_0^4 e^{2x} - x^2 dx$$

1 point is earned for the correct antidifferentiation and evaluation

0/1 if student doesn't have correct cone volume or doesn't indicate using cone

$$= \pi \left(\frac{1}{2} e^{2x} - \frac{x^3}{3} \right) \Big|_0^4$$

$$= \pi \left[\left(\frac{1}{2} e^8 - \frac{64}{3} \right) - \left(\frac{1}{2} e^0 - 0 \right) \right]$$

$$= \pi \left(\frac{1}{2} e^8 - \frac{131}{6} \right)$$

$$\approx 1468.646\pi \approx 4613.886$$

OR

Using y-axis 4/4:

3 points are earned for the correct integral where one points is earned for 2π and correct limits and two points are earned for the correct integrands

<1> each error

Note: 0/2 if any limits incorrect

$$2\pi \left[\int_0^1 y \cdot y \, dy + \int_1^4 y(y - \ln y) \, dy + \int_4^{e^4} y(4 - \ln y) \, dy \right]$$

1 point is earned for the correct answer

$$\approx 1468.646\pi \approx 4618.886$$

$$\text{Note: } \pi \left[\int_0^k e^{2x} - x^2 dx + \int_k^4 x^2 - e^{2x} dx \right]$$

Scores 1/3 for integral, and is eligible for fourth point



0	1	2	3	4
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The student response earns all of the following points:

Using Geometry (Volume of Cone) 4/4:

8-11-12

3 points are earned for the correct integral where one point is earned for π and correct limits and two points are earned for correctly using $(e^x)^2 - (x)^2$

$$<1> (e^x - x)^2$$

$$<1> e^{x^2} - x^2$$

$$<1> (x)^2 - (e^x)^2$$

$$<1> e^{x^2}$$

$$\begin{aligned} V_x &= \pi \int_0^4 (e^x)^2 - (x)^2 dx \\ &= \pi \int_0^4 e^{2x} - x^2 dx \end{aligned}$$

1 point is earned for the correct antidifferentiation and evaluation

0/1 if student doesn't have correct cone volume or doesn't indicate using cone

$$\begin{aligned} &= \pi \left(\frac{1}{2} e^{2x} - \frac{x^3}{3} \right) \Big|_0^4 \\ &= \pi \left[\left(\frac{1}{2} e^8 - \frac{64}{3} \right) - \left(\frac{1}{2} e^0 - 0 \right) \right] \\ &= \pi \left(\frac{1}{2} e^8 - \frac{131}{6} \right) \\ &\approx 1468.646\pi \approx 4613.886 \end{aligned}$$

OR

Using y-axis 4/4:

3 points are earned for the correct integral where one points is earned for 2π and correct limits and two points are earned for the correct integrands

<1> each error

Note: 0/2 if any limits incorrect

$$2\pi \left[\int_0^1 y \cdot y \, dy + \int_1^4 y(y - \ln y) \, dy + \int_4^{e^4} y(4 - \ln y) \, dy \right]$$

1 point is earned for the correct answer

$$\approx 1468.646\pi \approx 4618.886$$

$$\text{Note: } \pi \left[\int_0^k e^{2x} - x^2 \, dx + \int_k^4 x^2 - e^{2x} \, dx \right]$$

Scores 1/3 for integral, and is eligible for fourth point

Part C

1 point is earned for 2π and correct limits

1 point is earned for integrand

$$V_y = 2\pi \int_0^4 x(e^x - x) \, dx$$

OR

8-11-12

1 point is earned for $\pi \int_0^1 f(y) + \int_1^4 g(y) + \int_4^{e^4} h(y)$

1 point is earned for all three correct integrands

$$V_y = \pi \left[\int_0^1 y^2 dy + \int_1^4 y^2 - (\ln y)^2 dy + \int_4^{e^4} 16 - (\ln y)^2 dy \right]$$

Note: $2\pi \left[\int_0^k x(e^x - x)dx + \int_k^4 x(x - e^x)dx \right]$

Scores 1/2



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for 2π and correct limits

1 point is earned for integrand

$$V_y = 2\pi \int_0^4 x(e^x - x)dx$$

OR

1 point is earned for $\pi \int_0^1 f(y) + \int_1^4 g(y) + \int_4^{e^4} h(y)$

1 point is earned for all three correct integrands

$$V_y = \pi \left[\int_0^1 y^2 dy + \int_1^4 y^2 - (\ln y)^2 dy + \int_4^{e^4} 16 - (\ln y)^2 dy \right]$$

Note: $2\pi \left[\int_0^k x(e^x - x)dx + \int_k^4 x(x - e^x)dx \right]$

Scores 1/2

8-11-12

12. Let R be the region enclosed by the graphs of $y=e^x$, $y=x$ and the lines $x=0$ and $x=4$.

Find the volume of the solid generated when R is revolved about the x -axis.

Part B

Using Geometry (Volume of Cone) 4/4:

3 points are earned for the correct integral where one point is earned for π and correct limits and two points are earned for correctly using $(e^x)^2 - (x)^2$

$$< -1 > (e^x - x)^2$$

$$< -1 > e^{x^2} - x^2$$

$$< -1 > (x)^2 - (e^x)^2$$

$$< -1 > e^{x^2}$$

$$V_x = \pi \int_0^4 (e^x)^2 - (x)^2 dx$$

$$= \pi \int_0^4 e^{2x} - x^2 dx$$

1 point is earned for the correct antidifferentiation and evaluation

0/1 if student doesn't have correct cone volume or doesn't indicate using cone

$$\begin{aligned} &= \pi \left(\frac{1}{2} e^{2x} - \frac{x^3}{3} \right) \Big|_0^4 \\ &= \pi \left[\left(\frac{1}{2} e^8 - \frac{64}{3} \right) - \left(\frac{1}{2} e^0 - 0 \right) \right] \\ &= \pi \left(\frac{1}{2} e^8 - \frac{131}{6} \right) \\ &\approx 1468.646\pi \approx 4613.886 \end{aligned}$$

OR

Using y -axis 4/4:

3 points are earned for the correct integral where one point is earned for 2π and correct limits and two points are earned for the correct integrands

< -1 > each error

Note: 0/2 if any limits incorrect

$$2\pi \left[\int_0^1 y \cdot y dy + \int_1^4 y(y - \ln y) dy + \int_4^{e^4} y(4 - \ln y) dy \right]$$

1 point is earned for the correct answer

$$\approx 1468.646\pi \approx 4618.886$$

$$\text{Note: } \pi \left[\int_0^k e^{2x} - x^2 dx + \int_k^4 x^2 - e^{2x} dx \right]$$

Scores 1/3 for integral, and is eligible for fourth point



0	1	2	3	4
---	---	---	---	---

The student response earns four of the following points:

Using Geometry (Volume of Cone) 4/4:

3 points are earned for the correct integral where one point is earned for π and correct limits and two points are earned for correctly using $(e^x)^2 - (x)^2$

$$< -1 > (e^x - x)^2$$

$$< -1 > e^{x^2} - x^2$$

$$< -1 > (x)^2 - (e^x)^2$$

8-11-12

$$< -1 > e^{x^2}$$

$$V_x = \pi \int_0^4 (e^x)^2 - (x)^2 dx$$

$$= \pi \int_0^4 e^{2x} - x^2 dx$$

1 point is earned for the correct antidifferentiation and evaluation

0/1 if student doesn't have correct cone volume or doesn't indicate using cone

$$\begin{aligned} &= \pi \left(\frac{1}{2} e^{2x} - \frac{x^3}{3} \right) \Big|_0^4 \\ &= \pi \left[\left(\frac{1}{2} e^8 - \frac{64}{3} \right) - \left(\frac{1}{2} e^0 - 0 \right) \right] \\ &= \pi \left(\frac{1}{2} e^8 - \frac{131}{6} \right) \\ &\approx 1468.646\pi \approx 4613.886 \end{aligned}$$

OR

Using y-axis 4/4:

3 points are earned for the correct integral where one points is earned for 2π and correct limits and two points are earned for the correct integrands

< -1 > each error

Note: 0/2 if any limits incorrect

$$2\pi \left[\int_0^1 y \cdot y dy + \int_1^4 y(y - \ln y) dy + \int_4^{e^4} y(4 - \ln y) dy \right]$$

1 point is earned for the correct answer

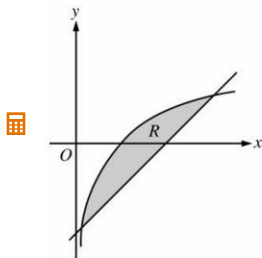
$$\approx 1468.646\pi \approx 4613.886$$

$$\text{Note: } \pi \left[\int_0^k e^{2x} - x^2 dx + \int_k^4 x^2 - e^{2x} dx \right]$$

Scores 1/3 for integral, and is eligible for fourth point

8-11-12

13.



Let R be the shaded region bounded by the graph of $y = \ln x$ and the line $y = x - 2$, as shown above.

- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is rotated about the horizontal line $y = -3$.
- (c) Write, but do not evaluate, an integral expression that can be used to find the volume of the solid generated when R is rotated about the y -axis.

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$\ln(x) = x - 2$ when $x = 0.15859$ and 3.14619 .

Let $S = 0.15859$ and $T = 3.14619$

$$\text{Area of } R = \int_S^T (\ln(x) - (x - 2)) dx = 1.949$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$\ln(x) = x - 2$ when $x = 0.15859$ and 3.14619 .

Let $S = 0.15859$ and $T = 3.14619$

$$\text{Area of } R = \int_S^T (\ln(x) - (x - 2)) dx = 1.949$$

Part B

2 points are earned for the integrand

1 point is earned for the limits, constant, and answer

$$\begin{aligned} \text{Volume} &= \pi \int_S^T ((\ln(x) + 3)^2 - (x - 2 + 3)^2) dx \\ &= 34.198 \text{ or } 34.199 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

8-11-12

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the limits, constant, and answer

$$\begin{aligned}\text{Volume} &= \pi \int_S^T \left((\ln(x) + 3)^2 - (x - 2 + 3)^2 \right) dx \\ &= 34.198 \text{ or } 34.199\end{aligned}$$

Part C

2 points are earned for the integrand

1 point is earned for the limits and constant

$$\text{Volume} = \pi \int_{S-2}^{T-2} \left((y+2)^2 - (e^y)^2 \right) dy$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the limits and constant

$$\text{Volume} = \pi \int_{S-2}^{T-2} \left((y+2)^2 - (e^y)^2 \right) dy$$

8-11-12

14.  Let f and g be the functions given by $f(x) = e^x$ and $g(x) = \ln x$.
- (a) Find the area of the region enclosed by the graphs of f and g between $x = \frac{1}{2}$ and $x = 1$.
- (b) Find the volume of the solid generated when the region enclosed by the graphs of f and g between $x = \frac{1}{2}$ and $x = 1$ is revolved about the line $y = 4$.
- (c) Let h be the function given by $h(x) = f(x) - g(x)$. Find the absolute minimum value of $h(x)$ on the closed interval $\frac{1}{2} \leq x \leq 1$, and find the absolute maximum value of $h(x)$ on the closed interval $\frac{1}{2} \leq x \leq 1$. Show the analysis that leads to your answers.

Part A

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_{1/2}^1 (e^x - \ln x) dx = 1.222 \text{ or } 1.223$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

$$\text{Area} = \int_{1/2}^1 (e^x - \ln x) dx = 1.222 \text{ or } 1.223$$

Part B

1 point is earned for limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_{1/2}^1 \left((4 - \ln x)^2 - (4 - e^x)^2 \right) dx \\ &= 7.515\pi \text{ or } 23.609 \end{aligned}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_{1/2}^1 \left((4 - \ln x)^2 - (4 - e^x)^2 \right) dx \\ &= 7.515\pi \text{ or } 23.609 \end{aligned}$$

Part C

8-11-12

1 point is earned for considering $h'(x) = 0$

1 point is earned for identifying critical point and endpoints as candidates

1 point is earned for the answers

$$h'(x) = f'(x) - g'(x) = e^x - \frac{1}{x} = 0$$

$$x = 0.567143$$

Absolute minimum value and absolute maximum value occur at the critical point or at the endpoints.

$$h(0.567143) = 2.330$$

$$h(0.5) = 2.3418$$

$$h(1) = 2.718$$

The absolute minimum is 2.330.

The absolute maximum is 2.718.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for considering $h'(x) = 0$

1 point is earned for identifying critical point and endpoints as candidates

1 point is earned for the answers

$$h'(x) = f'(x) - g'(x) = e^x - \frac{1}{x} = 0$$

$$x = 0.567143$$

Absolute minimum value and absolute maximum value occur at the critical point or at the endpoints.

$$h(0.567143) = 2.330$$

$$h(0.5) = 2.3418$$

$$h(1) = 2.718$$

The absolute minimum is 2.330.

The absolute maximum is 2.718.

8-11-12

15. Let f and g be the functions given by $f(x) = e^x$ and $g(x) = \ln x$.

 Find the volume of the solid generated when the region enclosed by the graphs of f and g between $x = \frac{1}{2}$ and $x = 1$ is revolved about the line $y = 4$.

Part B

1 point is earned for limits and constant

2 points are earned for integrand

< -1 > each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_{1/2}^1 \left((4 - \ln x)^2 - (4 - e^x)^2 \right) dx \\ &= 7.515\pi \text{ or } 23.609 \end{aligned}$$



0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for limits and constant

2 points are earned for integrand

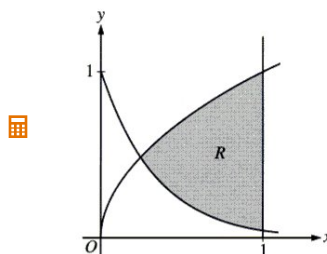
< -1 > each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_{1/2}^1 \left((4 - \ln x)^2 - (4 - e^x)^2 \right) dx \\ &= 7.515\pi \text{ or } 23.609 \end{aligned}$$

8-11-12

16.



Let R be the shaded region bounded by the graphs of $y = \sqrt{x}$ and $y = e^{-3x}$ and the vertical line $x = 1$, as shown in the figure above.

(a) Find the area of R .

(b) Find the volume of the solid generated when R is revolved about the horizontal line $y = 1$.

(c) The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a rectangle whose height is 5 times the length of its base in region R . Find the volume of this solid.

Part A

1 point is earned for the correct integrand

$$\text{Area} = \int_T^1 (\sqrt{x} - e^{-3x}) dx$$

1 point is earned for the correct answer

$$= 0.442 \text{ or } 0.443$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct integrand

$$\text{Area} = \int_T^1 (\sqrt{x} - e^{-3x}) dx$$

1 point is earned for the correct answer

$$= 0.442 \text{ or } 0.443$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

Part B

2 points are earned for the correct integrand

<-1> reversal

<-1> error with constant

<-1> omits 1 in one radius

8-11-12

<2> other errors

$$\text{Volume} = \pi \int_1^T ((1 - e^{-3x})^2 - (1 - \sqrt{x})^2) dx$$

1 point is earned for the correct answer

$$= 0.453\pi \text{ or } 1.423 \text{ or } 1.424$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct integrand

<-1> reversal

<-1> error with constant

<-1> omits 1 in one radius

<-2> other errors

$$\text{Volume} = \pi \int_1^T ((1 - e^{-3x})^2 - (1 - \sqrt{x})^2) dx$$

1 point is earned for the correct answer

$$= 0.453\pi \text{ or } 1.423 \text{ or } 1.424$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

Part C

2 points are earned for the correct integrand

<-1> incorrect but has $\sqrt{x} - e^{-3x}$ as a factor

$$\text{Length} = \sqrt{x} - e^{-3x}$$

$$\text{Height} = 5(\sqrt{x} - e^{-3x})$$

1 point is earned for the correct answer

$$\text{Volume} = \int_T^1 5(\sqrt{x} - e^{-3x})^2 dx = 1.554$$

1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

8-11-12

2 points are earned for the correct integrand

<-1> incorrect but has $\sqrt{x} - e^{-3x}$ as a factor

$$\text{Length} = \sqrt{x} - e^{-3x}$$

$$\text{Height} = 5(\sqrt{x} - e^{-3x})$$

1 point is earned for the correct answer

$$\text{Volume} = \int_T^1 5(\sqrt{x} - e^{-3x})^2 dx = 1.554$$

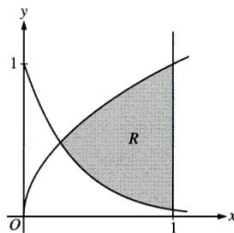
1 point is earned for correct limits in an integral in (a), (b), (c).

Points of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

8-11-12

17.



Let R be the shaded region bounded by the graphs of $y = \sqrt{x}$ and $y = e^{-3x}$ and the vertical line $x = 1$, as shown in the figure above.

Find the volume of the solid generated when R is revolved about the horizontal line $y = 1$.

Part B

2 points are earned for the correct integrand

< -1 > reversal

< -1 > error with constant

< -1 > omits 1 in one radius

< -2 > other errors

$$\text{Volume} = \pi \int_T^1 ((1 - e^{-3x})^2 - (1 - \sqrt{x})^2) dx$$

1 point is earned for the correct answer

$$= 0.453\pi \text{ or } 1.423 \text{ or } 1.424$$

✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the correct integrand

< -1 > reversal

< -1 > error with constant

< -1 > omits 1 in one radius

< -2 > other errors

$$\text{Volume} = \pi \int_T^1 ((1 - e^{-3x})^2 - (1 - \sqrt{x})^2) dx$$

1 point is earned for the correct answer

$$= 0.453\pi \text{ or } 1.423 \text{ or } 1.424$$

Overall

1 point is earned for for the correct limits in an integral in (a), (b), or (c)

Point of intersection

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

✓

0	1
---	---

The student response earns one of the following points:

1 point is earned for for the correct limits in an integral in (a), (b), or (c)

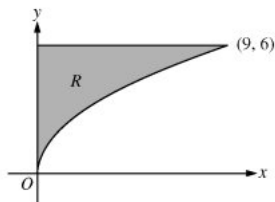
Point of intersection

8-11-12

$$e^{-3x} = \sqrt{x} \text{ at } (T, S) = (0.238734, 0.488604)$$

8-11-12

18.



Let R be the region in the first quadrant bounded by the graph of $y = 2\sqrt{x}$, the horizontal line $y = 6$, and the y -axis, as shown in the figure above.

(a) Find the area of R .

(b) Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 7$.

(c) Region R is the base of a solid. For each y , where $0 \leq y \leq 6$, the cross section of the solid taken perpendicular to the y -axis is a rectangle whose height is 3 times the length of its base in region R . Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part A

1 point is earned for integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^9 (6 - 2\sqrt{x}) dx = \left(6x - \frac{4}{3}x^{3/2} \right) \Big|_{x=0}^{x=9} = 18$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for the antiderivative

1 point is earned for the answer

$$\text{Area} = \int_0^9 (6 - 2\sqrt{x}) dx = \left(6x - \frac{4}{3}x^{3/2} \right) \Big|_{x=0}^{x=9} = 18$$

Part B

2 points are earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \pi \int_0^9 \left((7 - 2\sqrt{x})^2 - (7 - 6)^2 \right) dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \pi \int_0^9 \left((7 - 2\sqrt{x})^2 - (7 - 6)^2 \right) dx$$

8-11-12

Part C

2 points are earned for integrand
 1 point is earned for the answer

Solving $y = 2\sqrt{x}$ for x yields $x = \frac{y^2}{4}$.

Each rectangular cross section has area $\left(3\frac{y^2}{4}\right)\left(\frac{y^2}{4}\right) = \frac{3}{16}y^4$.

Volume = $\int_0^6 \frac{3}{16}y^4 dy$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:
 2 points are earned for integrand
 1 point is earned for the answer

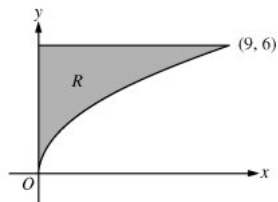
Solving $y = 2\sqrt{x}$ for x yields $x = \frac{y^2}{4}$.

Each rectangular cross section has area $\left(3\frac{y^2}{4}\right)\left(\frac{y^2}{4}\right) = \frac{3}{16}y^4$.

Volume = $\int_0^6 \frac{3}{16}y^4 dy$

8-11-12

19.



Let R be the region in the first quadrant bounded by the graph of $y = 2\sqrt{x}$, the horizontal line $y = 6$, and the y -axis, as shown in the figure above.

Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 7$.

Part B

2 point is earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \pi \int_0^9 \left((7 - 2\sqrt{x})^2 - (7 - 6)^2 \right) dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

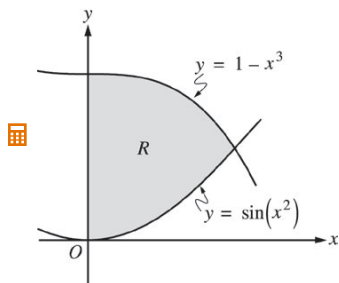
2 point is earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \pi \int_0^9 \left((7 - 2\sqrt{x})^2 - (7 - 6)^2 \right) dx$$

8-11-12

20.



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$, as shown in the figure above.

(a) Find the area of R .

(b) A horizontal line, $y = k$, is drawn through the point where the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect. Find k and determine whether this line divides R into two regions of equal area. Show the work that leads to your conclusion.

(c) Find the volume of the solid generated when R is revolved about the line $y = -3$.

Part A

1 point: For the integrand

1 point: For the correct answer

$$\begin{aligned}\text{Area} &= \int_0^A (1 - x^3 - \sin(x^2)) dx \\ &= 0.533 \text{ (or } 0.534\text{)}\end{aligned}$$

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

✓

0	1	2	3
---	---	---	---

The response earns all of the following points:

1 point: For the integrand

1 point: For the correct answer

$$\begin{aligned}\text{Area} &= \int_0^A (1 - x^3 - \sin(x^2)) dx \\ &= 0.533 \text{ (or } 0.534\text{)}\end{aligned}$$

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

Part B

1 point: For integral(s) with k value

1 point: For value(s) of integral(s)

$$k = B = 0.552352$$

$$\int_0^A (1 - x^3 - k) dx = 0.257 \text{ (or } 0.256\text{)}$$

$$\int_0^A (k - \sin(x^2)) dx = 0.277 \text{ (or } 0.276\text{)}$$

1 point: For conclusion tied to part (a) or comparison of two integrals

The two regions have unequal areas.

8-11-12

Note: Stating value of k does not earn a point.

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

✓

0	1	2	3	4
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The response earns all of the following points:

1 point: For integral(s) with k value

1 point: For value(s) of integral(s)

$$k = B = 0.552352$$

$$\int_0^A (1 - x^3 - k) dx = 0.257 \text{ (or } 0.256)$$

$$\int_0^A (k - \sin(x^2)) dx = 0.277 \text{ (or } 0.276)$$

1 point: For conclusion tied to part (a) or comparison of two integrals

The two regions have unequal areas.

Note: Stating value of k does not earn a point.

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

Part C

2 points: For the correct integrand

1 point: For the correct answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((1 - x^3 + 3)^2 - (\sin(x^2) + 3)^2 \right) dx \\ &= 11.841 \text{ (or } 11.840) \end{aligned}$$

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

✓

0	1	2	3	4
---	---	---	---	---

The response earns all of the following points:

2 points: For the correct integrand

1 point: For the correct answer

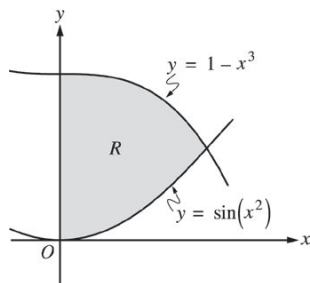
$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((1 - x^3 + 3)^2 - (\sin(x^2) + 3)^2 \right) dx \\ &= 11.841 \text{ (or } 11.840) \end{aligned}$$

1 point : For correct limits in an integral in (a), (b), or (c)

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$.

8-11-12

21.



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$, as shown in the figure above.

Find the volume of the solid generated when R is revolved about the line $y = -3$.

Part C

The response can earn up to 3 points:

2 points: For the correct integrand

1 point: For the correct answer

$$\text{Volume} = \pi \int_0^A (1 - x^3 + 3)^2 - (\sin(x^2) + 3)^2 dx$$

$$\text{Volume} = 11.841 \text{ (or } 11.840\text{)}$$

✓

0	1	2	3
---	---	---	---

The response earns all three of the following points:

2 points: For the correct integrand

1 point: For the correct answer

$$\text{Volume} = \pi \int_0^A (1 - x^3 + 3)^2 - (\sin(x^2) + 3)^2 dx$$

$$\text{Volume} = 11.841 \text{ (or } 11.840\text{)}$$

Limits

Can earn this point only once in the overall scenario:

1 point : For correct limits in the integral

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$

✓

0	1
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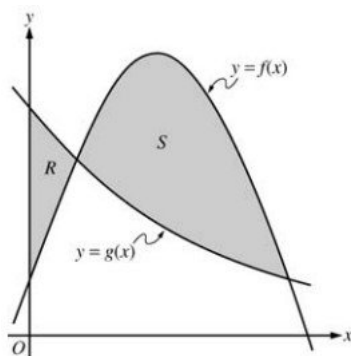
The response earns the following point:

1 point : For correct limits in the integral

The graphs of $y = 1 - x^3$ and $y = \sin(x^2)$ intersect in the first quadrant at the point $(A, B) = (0.764972, 0.552352)$

8-11-12

22.



Let f and g be the functions given by $f(x) = \frac{1}{4} + \sin(\pi x)$ and $g(x) = 4^{-x}$. Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of f and g , and let S be the shaded region in the first quadrant enclosed by the graphs of f and g , as shown in the figure above.

- Find the area of R .
- Find the area of S .
- Find the volume of the solid generated when S is revolved about the horizontal line $y = -1$.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$f(x) = g(x) \text{ when } \frac{1}{4} + \sin(\pi x) = 4^{-x}.$$

f and g intersect when $x = 0.178218$ and when $x = 1$.

Let $a = 0.178218$

$$\int_0^a (g(x) - f(x)) dx = 0.064 \text{ or } 0.065$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$f(x) = g(x) \text{ when } \frac{1}{4} + \sin(\pi x) = 4^{-x}.$$

f and g intersect when $x = 0.178218$ and when $x = 1$.

Let $a = 0.178218$

$$\int_0^a (g(x) - f(x)) dx = 0.064 \text{ or } 0.065$$

Part B

1 point is earned for the limits

8-11-12

1 point is earned for the integrand

1 point is earned for the answer

$$\int_a^1 (f(x) - g(x))dx = 0.410$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_a^1 (f(x) - g(x))dx = 0.410$$

Part C

2 points are earned for the integrand

1 point is earned for the limits, constant, and answer

$$\pi \int_a^1 ((f(x) + 1)^2 - (g(x) + 1)^2)dx = 4.558 \text{ or } 4.559$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

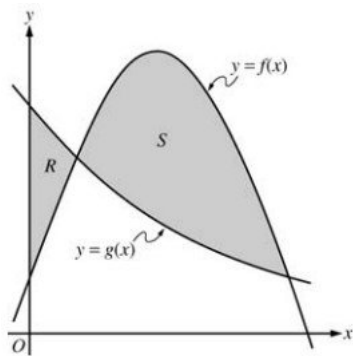
2 points are earned for the integrand

1 point is earned for the limits, constant, and answer

$$\pi \int_a^1 ((f(x) + 1)^2 - (g(x) + 1)^2)dx = 4.558 \text{ or } 4.559$$

8-11-12

23.



Let f and g be the functions given by $f(x) = \frac{1}{4} + \sin(\pi x)$ and $g(x) = 4^{-x}$. Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of f and g , and let S be the shaded region in the first quadrant enclosed by the graphs of f and g , as shown in the figure above.

Find the volume of the solid generated when S is revolved about the horizontal line $y = -1$.

Part C

2 points are earned for the integrand

1 point is earned for the limits, constant, and answer

$$\pi \int_a^1 \left((f(x) + 1)^2 - (g(x) + 1)^2 \right) dx = 4.558 \text{ or } 4.559$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

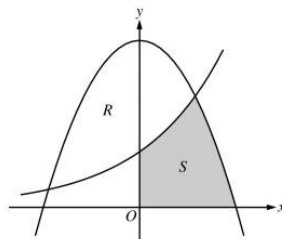
2 points are earned for the integrand

1 point is earned for the limits, constant, and answer

$$\pi \int_a^1 \left((f(x) + 1)^2 - (g(x) + 1)^2 \right) dx = 4.558 \text{ or } 4.559$$

8-11-12

24.



Let R and S in the figure above be defined as follows: R is the region in the first and second quadrants bounded by the graphs of $y = 3 - x^2$ and $y = 2^x$. S is the shaded region in the first quadrant bounded by the two graphs, the x -axis, and the y -axis.

- Find the area of S .
- Find the volume of the solid generated when R is rotated about the horizontal line $y = -1$.
- The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an isosceles right triangle with one leg across the base of the solid. Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part A

1 point is earned for integrands

1 point is earned for limits

1 point is earned for the answer

$$3 - x^2 = 2^x \text{ when } x = -1.63658 \text{ and } x = 1$$

$$\text{Let } a = -1.63658$$

$$\begin{aligned} \text{Area of } S &= \int_0^1 2^x dx + \int_1^{\sqrt{3}} (3 - x^2) dx \\ &= 2.240 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrands

1 point is earned for limits

1 point is earned for the answer

$$3 - x^2 = 2^x \text{ when } x = -1.63658 \text{ and } x = 1$$

$$\text{Let } a = -1.63658$$

$$\begin{aligned} \text{Area of } S &= \int_0^1 2^x dx + \int_1^{\sqrt{3}} (3 - x^2) dx \\ &= 2.240 \end{aligned}$$

Part B

2 point are earned for integrand

1 point is earned for limits and constant

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_a^1 \left((3 - x^2 + 1)^2 - (2^x + 1)^2 \right) dx \\ &= 63.106 \text{ or } 63.107 \end{aligned}$$

8-11-12

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 point are earned for integrand

1 point is earned for limits and constant

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_a^1 \left((3 - x^2 + 1)^2 - (2^x + 1)^2 \right) dx \\ &= 63.106 \text{ or } 63.107\end{aligned}$$

Part C

1 point is earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \frac{1}{2} \int_a^1 (3 - x^2 - 2^x)^2 dx$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

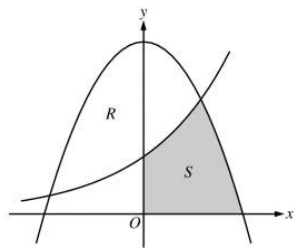
1 point is earned for integrand

1 point is earned for limits and constant

$$\text{Volume} = \frac{1}{2} \int_a^1 (3 - x^2 - 2^x)^2 dx$$

8-11-12

25.



Let R and S in the figure above be defined as follows: R is the region in the first and second quadrants bounded by the graphs of $y = 3 - x^2$ and $y = 2^x$. S is the shaded region in the first quadrant bounded by the two graphs, the x -axis, and the y -axis.

Find the volume of the solid generated when R is rotated about the horizontal line $y = -1$.

Part B

2 point are earned for integrand

1 point is earned for limits and constant

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_{-1}^1 \left((3 - x^2 + 1)^2 - (2^x + 1)^2 \right) dx \\ &= 63.106, \text{ or }, 63.107 \end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 point are earned for integrand

1 point is earned for limits and constant

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_{-1}^1 \left((3 - x^2 + 1)^2 - (2^x + 1)^2 \right) dx \\ &= 63.106, \text{ or }, 63.107 \end{aligned}$$

8-11-12

26. Let R be the region between the graphs of $y = 1 + \sin(\pi x)$ and $y = x^2$ from $x = 0$ to $x = 1$.

Set up, but do not integrate an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the x -axis.

Part B

2 points are earned for integral expression

1 point is deducted for incorrect limits

$$V = \pi \int_0^1 (1 + \sin(\pi x))^2 - x^4 dx$$

or

$$2\pi \int_0^1 y^{3/2} dy + 2\pi \int_1^2 y \left(1 - \frac{2}{\pi} \arcsin(y - 1)\right) dy$$

✓

0	1	2
---	---	---

The student response earns two of the following points:

2 points are earned for integral expression

1 point is deducted for incorrect limits

$$V = \pi \int_0^1 (1 + \sin(\pi x))^2 - x^4 dx$$

or

$$2\pi \int_0^1 y^{3/2} dy + 2\pi \int_1^2 y \left(1 - \frac{2}{\pi} \arcsin(y - 1)\right) dy$$

8-11-12

27. Let R be the region in the first quadrant bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$.
 Find the volume of the solid generated when R is rotated about the vertical line $x = -1$.

Part B

1 point is earned for constant and limits

2 points are earned for integrand

1 point is earned for the answer

$$\pi \int_0^3 \left((3y+1)^2 - (y^2+1)^2 \right) dy$$

$$= \frac{207\pi}{5} = 130.061 \text{ or } 130.062$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for constant and limits

2 points are earned for integrand

1 point is earned for the answer

$$\pi \int_0^3 \left((3y+1)^2 - (y^2+1)^2 \right) dy$$

$$= \frac{207\pi}{5} = 130.061 \text{ or } 130.062$$

28. Let S be the region enclosed by the graphs of $y = 2x$ and $y = 2x^2$ for $0 \leq x \leq 1$. What is the volume of the solid generated when S is revolved about the line $y = 3$?

(A) $\pi \int_0^1 \left((3 - 2x^2)^2 - (3 - 2x)^2 \right) dx$

(B) $\pi \int_0^1 \left((3 - 2x)^2 - (3 - 2x^2)^2 \right) dx$

(C) $\pi \int_0^1 (4x^2 - 4x^4) dx$

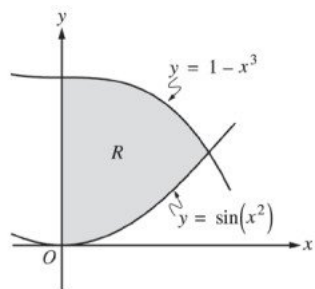
(D) $\pi \int_0^2 \left(\left(3 - \frac{y}{2} \right)^2 - \left(3 - \sqrt{\frac{y}{2}} \right)^2 \right) dy$

(E) $\pi \int_0^2 \left(\left(3 - \sqrt{\frac{y}{2}} \right)^2 - \left(3 - \frac{y}{2} \right)^2 \right) dy$



8-11-12

29.



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = 1 - x^3$ and $y = \sin(x^2)$, as shown in the figure above.

Find the volume of the solid generated when R is revolved about the line $y = -3$.

Part C

Two points are earned for the integrand

One point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left((1 - x^3 + 3)^2 - (\sin(x^2) + 3)^2 \right) dx \\ &= 11.841(\text{or } 11.840)\end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student earns all of the following points:

Two points are earned for the integrand

One point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left((1 - x^3 + 3)^2 - (\sin(x^2) + 3)^2 \right) dx \\ &= 11.841(\text{or } 11.840)\end{aligned}$$

8-11-12

30. Let R be the region in the first quadrant enclosed by the hyperbola $x^2 - y^2 = 9$, the x -axis, and the line $x = 5$.

Set up, but do not integrate, an integral expression in terms of a single variable for the volume of solid generated when R is revolved about the line $x = -1$.

Part B

3 points are earned for a correct integrand

1 point is earned for appropriate limit and $k\pi$

Shells:

$$\begin{aligned} V &= 2\pi \int_3^5 (x+1)y \, dx \\ &= 2\pi \int_3^5 (x+1)\sqrt{x^2-9} \, dx \end{aligned}$$

or Washers

$$\begin{aligned} V &= \pi \int_0^4 \left[(36 - (x+1)^2) \right] dy \\ &= \pi \int_0^4 \left[36 - (\sqrt{9-y^2} + 1)^2 \right] dy \end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

3 points are earned for a correct integrand

1 point is earned for appropriate limit and $k\pi$

Shells:

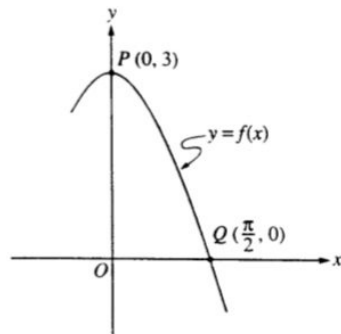
$$\begin{aligned} V &= 2\pi \int_3^5 (x+1)y \, dx \\ &= 2\pi \int_3^5 (x+1)\sqrt{x^2-9} \, dx \end{aligned}$$

or Washers

$$\begin{aligned} V &= \pi \int_0^4 \left[(36 - (x+1)^2) \right] dy \\ &= \pi \int_0^4 \left[36 - (\sqrt{9-y^2} + 1)^2 \right] dy \end{aligned}$$

8-11-12

31.



Let f be the function given by $f(x) = 3 \cos x$. As shown above, the graph of f crosses the y -axis at point P and the x -axis at point Q .

Let R be the region in the first quadrant bounded by the graph of f and line segment PQ . Write an integral expression for the volume of the solid generated by revolving the region R about the x -axis. Do not evaluate.

Part D

2 points are earned for the correct integrand

0/2 if not difference of 2 squares

1/2 if incorrect but of form $(a \cos x)^2 - (bx + c)^2$; $a, b \neq 0$

1/2 if reversed

1 point is earned for the correct constant and limits

0/3 for shell around y -axis

$$V = \pi \int_0^{\pi/2} \left[(3 \cos x)^2 - \left(-\frac{6}{\pi}x + 3 \right)^2 \right] dx$$

✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for the correct integrand

0/2 if not difference of 2 squares

1/2 if incorrect but of form $(a \cos x)^2 - (bx + c)^2$; $a, b \neq 0$

1/2 if reversed

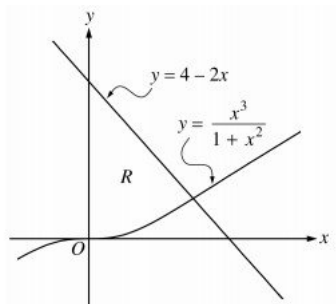
1 point is earned for the correct constant and limits

0/3 for shell around y -axis

$$V = \pi \int_0^{\pi/2} \left[(3 \cos x)^2 - \left(-\frac{6}{\pi}x + 3 \right)^2 \right] dx$$

8-11-12

32.



Let R be the region bounded by the y -axis and the graphs of $y = \frac{x^3}{1+x^2}$ and $y = 4 - 2x$, as shown in the figure above.

 Find the volume of the solid generated when R is revolved about the x -axis.

Part B

2 points are earned for the integrand and constant $< -1 >$ each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((4 - 2x)^2 - \left(\frac{x^3}{1 + x^2} \right)^2 \right) dx \\ &= 31.884 \text{ or } 31.885 \text{ or } 10.149\pi \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns two of the following points:

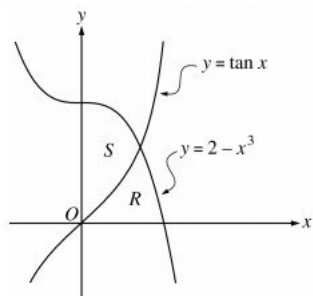
2 points are earned for the integrand and constant $< -1 >$ each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((4 - 2x)^2 - \left(\frac{x^3}{1 + x^2} \right)^2 \right) dx \\ &= 31.884 \text{ or } 31.885 \text{ or } 10.149\pi \end{aligned}$$

8-11-12

33.



Let R and S be the regions in the first quadrant shown in the figure above. The region R is bounded by the x -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$. The region S is bounded by the y -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$.



Find the volume of the solid generated when S is revolved about the x -axis.

Part C

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((2 - x^3)^2 - \tan^2 x \right) dx \\ &= 2.652\pi \text{ or } 8.331 \text{ or } 8.332 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((2 - x^3)^2 - \tan^2 x \right) dx \\ &= 2.652\pi \text{ or } 8.331 \text{ or } 8.332 \end{aligned}$$

8-11-12

34. Let R be the region enclosed by the graph of $y = \sqrt{x-1}$, the vertical line $x=10$, and the x -axis.



Find the volume of the solid generated when R is revolved about the horizontal line $y=3$

Part B

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_1^{10} (9 - (3 - \sqrt{x-1})^2) dx$$

$$= 212.057 \text{ or } 212.058$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_1^{10} (9 - (3 - \sqrt{x-1})^2) dx$$

$$= 212.057 \text{ or } 212.058$$

8-11-12

Let R be the region in the first quadrant enclosed by the graph of $y = \sqrt{6x + 4}$, the line $y = 2x$, and the y -axis.

35. Set up, but **do not integrate**, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the x -axis.

Part B

1 point is earned for integrand

1 point is earned for limits and constant

2 points **deducted** for incorrect integrand

1 point **deducted** for wrong limits, wrong constant or reversal

Volume about x -axis V

$$V = \pi \int_0^2 (6x + 4) - 4x^2 \, dx$$

or

$$V = \pi \int_0^2 (6x + 4) \, dx - \frac{32\pi}{3}$$

✓

0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for integrand

1 point is earned for limits and constant

1 point **deducted** for wrong limits, wrong constant or reversal

Volume about x -axis

$$V = \pi \int_0^2 (6x + 4) - 4x^2 \, dx$$

or

$$V = \pi \int_0^2 (6x + 4) \, dx - \frac{32\pi}{3}$$

8-11-12

36. Set up, but **do not integrate**, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the **y-axis**.

Part C

1 point is earned for integrand

1 point is earned for limits and constant

1 point **deducted** for wrong limits, wrong constant or reversal

Volume about y-axis V

$$V = 2\pi \int_0^2 x \left(\sqrt{6x+4} - 2x \right) dx$$

or

$$V = \pi \int_0^4 \frac{y^2}{2} dy - \pi \int_2^4 \left(\frac{y^2 - 4}{6} \right)^2 dy$$

✓

0	1	2
---	---	---

The student response earns two of the following points:

1 point is earned for integrand

1 point is earned for limits and constant

1 point **deducted** for wrong limits, wrong constant or reversal

Volume about y-axis

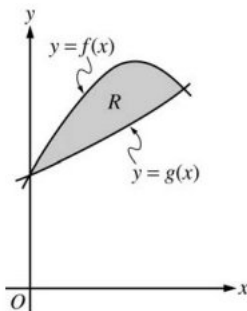
$$V = 2\pi \int_0^2 x \left(\sqrt{6x+4} - 2x \right) dx$$

or

$$V = \pi \int_0^4 \frac{y^2}{2} dy - \pi \int_2^4 \left(\frac{y^2 - 4}{6} \right)^2 dy$$

8-11-12

37.



Let f and g be the functions given by $f(x) = 1 + \sin(2x)$ and $g(x) = e^{x/2}$. Let R be the shaded region in the first quadrant enclosed by the graphs of f and g as shown in the figure above.

Find the volume of the solid generated when R is revolved about the x -axis.

Part B

The student response earns none of the following points:

2 point is earned for the integrand

(-1) for each error

Note: 0/2 if integral not of form $c \int_a^b \left((R^2(x) - (r^2(x))) dx \right)$

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^8 \left((f(x))^2 - (g(x))^2 \right) dx \\ &= \pi \int_0^8 \left((1 + \sin(2x))^2 - \left(e^{x/2} \right)^2 \right) dx \end{aligned}$$

=4.266 or 4.267

✓

0	1	2
---	---	---

The student response earns none of the following points:

2 point is earned for the integrand

(-1) for each error

Note: 0/2 if integral not of form $c \int_a^b \left((R^2(x) - (r^2(x))) dx \right)$

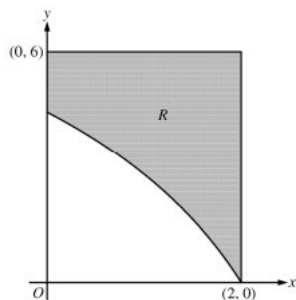
1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^8 \left((f(x))^2 - (g(x))^2 \right) dx \\ &= \pi \int_0^8 \left((1 + \sin(2x))^2 - \left(e^{x/2} \right)^2 \right) dx \end{aligned}$$

=4.266 or 4.267

8-11-12

38.



In the figure above, R is the shaded region in the first quadrant bounded by the graph of $y=4\ln(3-x)$, the horizontal line $y=6$, and the vertical line $x=2$.

Find the volume of the solid generated when R is revolved about the horizontal line $y=8$.

Part B

2 points are earned for the integrand

1 point is earned for the answer

$$\pi \int_0^2 \left((8 - 4\ln(3-x))^2 - (8-6)^2 \right) dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

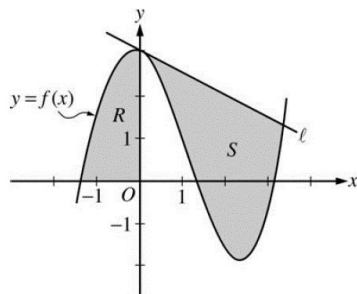
1 point is earned for the answer

$$\pi \int_0^2 \left((8 - 4\ln(3-x))^2 - (8-6)^2 \right) dx$$

=168.179 or 168.180

8-11-12

39.



Let f be the function given by $f(x) = \frac{x^3}{4} - \frac{x^2}{3} - \frac{x}{2} + 3 \cos x$. Let R be the shaded region in the second quadrant bounded by the graph of f , and let S be the shaded region bounded by the graph of f and line ℓ , the line tangent to the graph of f at $x = 0$, as shown above.

Find the volume of the solid generated when R is rotated about the horizontal line $y = -2$.

Part B

1 point is earned for limits and constant

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_{-1}^0 \left((f(x) + 2)^2 - 4 \right) dx = 59.361$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

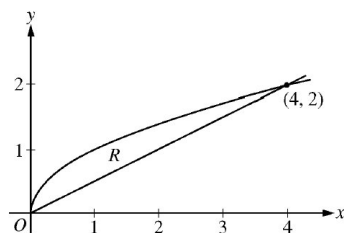
2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \int_{-1}^0 \left((f(x) + 2)^2 - 4 \right) dx = 59.361$$

8-11-12

40.



Let R be the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$, as shown in the figure above.

Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 2$.

Part C

The student response earns none of the following points:

1 point is earned for the limits and constant

2 points are earned for the integrand

$$Voulme = \pi \int_0^4 \left(\left(2 - \frac{x}{2} \right)^2 - (2 - \sqrt{x})^2 \right) dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

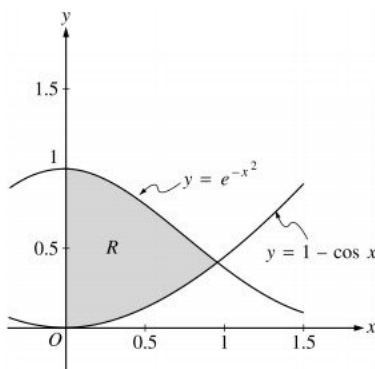
1 point is earned for the limits and constant

2 points are earned for the integrand

$$Voulme = \pi \int_0^4 \left(\left(2 - \frac{x}{2} \right)^2 - (2 - \sqrt{x})^2 \right) dx$$

8-11-12

41.



Let R be the shaded region in the first quadrant enclosed by the graphs of $y = e^{-x^2}$, $y = 1 - \cos x$, and the y -axis, as shown in the figure above.

Find the volume of the solid generated when the region R is revolved about the x -axis.

Part B

2 points are earned for the integrand and constant

< -1 > each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((e^{-x^2})^2 - (1 - \cos x)^2 \right) dx \\ &= 0.55596\pi = 1.746, \text{ or, } 1.747 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand and constant

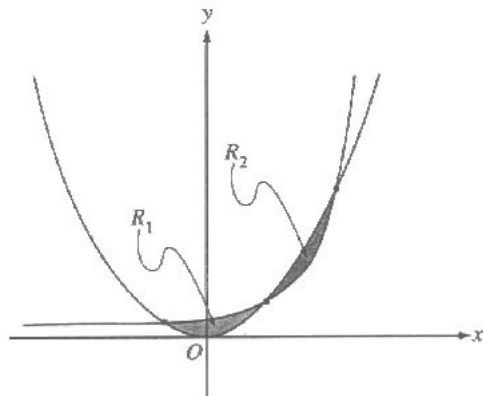
< -1 > each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((e^{-x^2})^2 - (1 - \cos x)^2 \right) dx \\ &= 0.55596\pi = 1.746, \text{ or, } 1.747 \end{aligned}$$

8-11-12

42.



Note: Figure not drawn to scale.

The shaded regions R_1 and R_2 shown above are enclosed by the graphs of $f(x)=x^2$ and $g(x)=2^x$.

Without using absolute value, set up an expression involving one or more integrals that gives the volume of the solid generated by revolving the region R_1 about the line $y = 5$. Do not evaluate.

Part C

1 point is earned for the correct limits and π .

2 points are earned for the correct integrand

< -1 > reversal

< -1 > uses absolute value

< -1 > with additional integral

$$\pi \int_{-0.767}^2 \left((5 - x^2)^2 - (5 - 2^x)^2 \right) dx$$

OR

1 point is earned for the correct limits and π .

2 points are earned for the correct integrand where one point is earned for each correct integrand

$$2\pi \int_0^{0.588} (5 - y)2\sqrt{y} dy + 2\pi \int_{0.588}^4 (5 - y) \left(\sqrt{y} - \frac{\ln y}{\ln 2} \right) dy$$

✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for the correct limits and π .

2 points are earned for the correct integrand

< -1 > reversal

< -1 > uses absolute value

< -1 > with additional integral

$$\pi \int_{-0.767}^2 \left((5 - x^2)^2 - (5 - 2^x)^2 \right) dx$$

OR

1 point is earned for the correct limits and π .

2 points are earned for the correct integrand where one point is earned for each correct integrand

8-11-12

$$2\pi \int_0^{0.588} (5-y)2\sqrt{y}dy + 2\pi \int_{0.588}^4 (5-y)\left(\sqrt{y} - \frac{\ln y}{\ln 2}\right)dy$$

43. Let f be the function given by $f(x)=e^{-x}$, and let g be the function given by $g(x)=kx$, where k is the nonzero constant such that the graph of f is tangent to the graph of g .

Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated by revolving the region R , given in part (b), about the x -axis.

Part C

2 point are earned for setting up integral with correct integrand

1 point **deducted** for incorrect limits/error involving π

$$\pi \int_{-1}^0 \left((e^{-x})^2 - (-ex)^2 \right) dx$$

✓

0	1	2
---	---	---

The student response earns two of the following points:

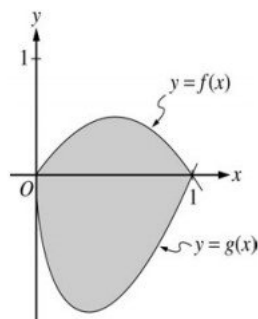
2 point are earned for setting up integral with correct integrand

1 point **deducted** for incorrect limits/error involving π

$$\pi \int_{-1}^0 \left((e^{-x})^2 - (-ex)^2 \right) dx$$

8-11-12

44.



Let f and g be the functions given by $f(x) = 2x(1 - x)$ and $g(x) = 3(x - 1)\sqrt{x}$ for $0 \leq x \leq 1$. The graphs of f and g are shown in the figure above.

Find the volume of the solid generated when the shaded region enclosed by the graphs of f and g is revolved above the horizontal line $y = 2$.

Part B

1 point is earned for the limits and constant

2 points are earned for integrand

< -1 > each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^1 \left((2 - g(x))^2 - (2 - f(x))^2 \right) dx \\ &= \pi \int_0^1 \left((2 - 3(x - 1)\sqrt{x})^2 - (2 - 2x(1 - x))^2 \right) dx \\ &= 16.179 \end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

2 points are earned for integrand

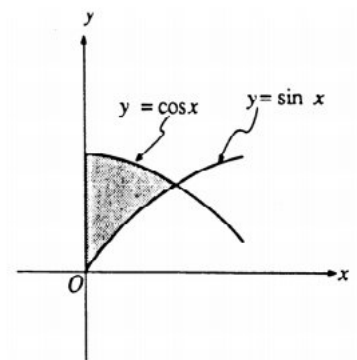
< -1 > each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^1 \left((2 - g(x))^2 - (2 - f(x))^2 \right) dx \\ &= \pi \int_0^1 \left((2 - 3(x - 1)\sqrt{x})^2 - (2 - 2x(1 - x))^2 \right) dx \\ &= 16.179 \end{aligned}$$

8-11-12

45.



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = \sin x$ and $y = \cos x$, as shown in the figure above. Find the volume of the solid generated when R is revolved about the x -axis.

Part B

1 point is earned for limits

1 point is earned for integrand

1 point is **deducted** for constant error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx \\
 &= \pi \int_0^{\pi/4} \cos 2x \, dx \\
 &= \frac{\pi}{2} \sin 2x \Big|_0^{\pi/4} \\
 &= \frac{\pi}{2} (1 - 0) = \frac{\pi}{2}
 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

1 point is earned for limits

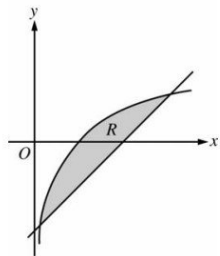
1 point is earned for integrand

1 point is **deducted** for constant error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx \\
 &= \pi \int_0^{\pi/4} \cos 2x \, dx \\
 &= \frac{\pi}{2} \sin 2x \Big|_0^{\pi/4} \\
 &= \frac{\pi}{2} (1 - 0) = \frac{\pi}{2}
 \end{aligned}$$

8-11-12



Let R be the shaded region bounded by the graph of $y = \ln x$ and the line $y = x - 2$, as show above.

46. Find the volume of the solid generated when R is rotated about the horizontal line $y = -3$.

Part B

2 points are earned for the integrand

1 point is earned for the limits

$$\text{Volume} = \pi \int_S^T \left((\ln(x) + 3)^2 - (x - 2 + 3)^2 \right) dx = 34.198 \text{ or } 34.199$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the limits

$$\text{Volume} = \pi \int_S^T \left((\ln(x) + 3)^2 - (x - 2 + 3)^2 \right) dx = 34.198 \text{ or } 34.199$$

47. Write, but do not evaluate, an integral expression that can be used to find the volume of the solid generated when R is rotated about the y -axis.

Part C

2 points are earned for the integrand

1 point is earned for the limits and constant

$$\text{Volume} = \pi \int_{S-2}^{T-2} \left((y + 2)^2 - (e^y)^2 \right) dy$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the limits and constant

$$\text{Volume} = \pi \int_{S-2}^{T-2} \left((y + 2)^2 - (e^y)^2 \right) dy$$

8-11-12

Let R be the region enclosed by the graphs of $y=e^x$, $y=(x-1)^2$, and the line $x=1$.

48. Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part C

2 points are earned for Definite integral

2 points are **deducted** for f,g, or u error

1 point is **deducted** for limit error, K error and integrand reversal

$$V = 2\pi \int_1^0 x \left[e^x - (x-1)^2 \right] dx$$

or

$$V = \pi \int_1^0 1 - (1 - \sqrt{y})^2 dy + \pi \int_e^1 1 - (\ln y)^2 dy$$

✓

0	1	2
---	---	---

The student response earns none of the following points:

2 points are earned for Definite integral

2 points are **deducted** for f,g, or u error

1 point is **deducted** for limit error, K error and integrand reversal

$$V = 2\pi \int_1^0 x \left[e^x - (x-1)^2 \right] dx$$

or

$$V = \pi \int_1^0 1 - (1 - \sqrt{y})^2 dy + \pi \int_e^1 1 - (\ln y)^2 dy$$

8-11-12

49. Find the volume of the solid generated when R is revolved about the x -axis.

Part B

2 points are earned for definite integral

2 points are **deducted** for faulty f , g , or u

1 point is earned for correct antiderivative

1 point is earned for evaluation of anti derivative

$$\begin{aligned} V &= \pi \int_0^1 e^{2x} - (x-1)^4 dx \\ &= \pi \left[\frac{e^{2x}}{2} \right]_0^1 - \pi \left[\frac{1}{5}(x-1)^5 \right]_0^1 \\ &= \pi \left[\left(\frac{e^2}{2} - \frac{1}{2} \right) - \frac{1}{5} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right) \end{aligned}$$

or

$$\begin{aligned} V &= 2\pi \int_0^1 y[1 - (1 - \sqrt{y})] dy + 2\pi \int_1^e y(1 - \ln y) dy \\ &= 2\pi \cdot \frac{2}{5} y^{5/2} \Big|_0^1 + 2\pi \left[\frac{1}{2} y^2 - \left(\frac{1}{2} y^2 \ln y - \frac{1}{4} y^2 \right) \right] \Big|_1^e \\ &= \frac{4}{5} \pi + 2\pi \left[\frac{1}{4} e^2 - \frac{3}{4} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right) \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

2 points are earned for definite integral

2 points are **deducted** for faulty f , g , or u

1 point is earned for correct antiderivative

1 point is earned for evaluation of anti derivative

$$\begin{aligned} V &= \pi \int_0^1 e^{2x} - (x-1)^4 dx \\ &= \pi \left[\frac{e^{2x}}{2} \right]_0^1 - \pi \left[\frac{1}{5}(x-1)^5 \right]_0^1 \\ &= \pi \left[\left(\frac{e^2}{2} - \frac{1}{2} \right) - \frac{1}{5} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right) \end{aligned}$$

or

$$\begin{aligned} V &= 2\pi \int_0^1 y[1 - (1 - \sqrt{y})] dy + 2\pi \int_1^e y(1 - \ln y) dy \\ &= 2\pi \cdot \frac{2}{5} y^{5/2} \Big|_0^1 + 2\pi \left[\frac{1}{2} y^2 - \left(\frac{1}{2} y^2 \ln y - \frac{1}{4} y^2 \right) \right] \Big|_1^e \\ &= \frac{4}{5} \pi + 2\pi \left[\frac{1}{4} e^2 - \frac{3}{4} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right) \end{aligned}$$

8-11-12

50. Consider the curve $y^2=4+x$ and chord AB joining the points $A(-4,0)$ and $B(0,2)$ on the curve.

Find the volume of the solid generated when the region R , defined in part (b), is revolved about the x -axis.

Part C

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0/1 if integral not of the form $\int_a^b [(f(x))^2 - (g(x))^2] dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\begin{aligned} & \pi \int_{-4}^0 \left((\sqrt{4+x})^2 - \left(\frac{1}{2}x + 2 \right)^2 \right) dx \\ &= \pi \int_{-4}^0 \left(4+x - \left(\frac{1}{4}x^2 + 2x + 4 \right) \right) dx \\ &= \pi \left(8 - \frac{16}{3} \right) = \frac{8\pi}{3} \approx 8.378 \end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0/1 if integral not of the form $\int_a^b [(f(x))^2 - (g(x))^2] dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\int_0^2 2\pi y [(2y-4) - (y^2-4)] dy = \frac{8\pi}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0/1 if integral not of the form $\int_a^b [(f(x))^2 - (g(x))^2] dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

8-11-12

$$\begin{aligned}\pi \int_{-4}^0 (\sqrt{4+x})^2 dx &= \pi \left(4x + \frac{x^2}{2} \right) \Big|_{-4}^0 \\ &= 0 - \pi(-16 + 8) = 8\pi \\ \text{Volume of cone} &= \frac{1}{3} \pi (2)^2 (4) = \frac{16\pi}{3} \\ \text{Volume} &= 8\pi - \frac{16\pi}{3} = \frac{8\pi}{3}\end{aligned}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

✓

0	1	2	3
---	---	---	---

The student response earns three of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0/1 if integral not of the form $\int_a^b [(f(x))^2 - (g(x))^2] dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\begin{aligned}\pi \int_{-4}^0 \left((\sqrt{4+x})^2 - \left(\frac{1}{2}x + 2 \right)^2 \right) dx \\ &= \pi \int_{-4}^0 \left(4 + x - \left(\frac{1}{4}x^2 + 2x + 4 \right) \right) dx \\ &= \pi \left(8 - \frac{16}{3} \right) = \frac{8\pi}{3} \approx 8.378\end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0/1 if integral not of the form $\int_a^b [(f(x))^2 - (g(x))^2] dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\int_0^2 2\pi y [(2y - 4) - (y^2 - 4)] dy = \frac{8\pi}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

8-11-12

1 point is earned for the correct antidifferentiation and evaluation

Note: 0/1 if integral not of the form $\int_a^b [(f(x))^2 - (g(x))^2] dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\pi \int_{-4}^0 (\sqrt{4+x})^2 dx = \pi \left(4x + \frac{x^2}{2} \right) \Big|_{-4}^0$$

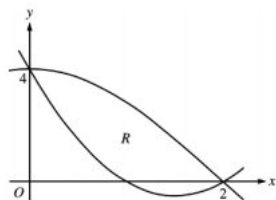
$$= 0 - \pi(-16 + 8) = 8\pi$$

$$\text{Volume of cone} = \frac{1}{3} \pi (2)^2 (4) = \frac{16\pi}{3}$$

$$\text{Volume} = 8\pi - \frac{16\pi}{3} = \frac{8\pi}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

51.



Let $f(x) = 2x^2 - 6x + 4$ and $g(x) = 4\cos(1/4 \pi x)$. Let R be the region bounded by the graphs of f and g , as shown in the figure above.

Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y=4$.

Part B

2 points are earned for the integrand.

1 point is earned for the limits and constant.

$$\text{Volume} = \pi \int_0^2 [(4 - f(x))^2 - (4 - g(x))^2] dx$$

$$\pi \int_0^2 \left[\left(4 - (2x^2 - 6x + 4) \right)^2 - \left(4 - 4\cos\left(\frac{\pi}{4}x\right) \right)^2 \right] dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the limits and constant

$$\text{Volume} = \pi \int_0^2 [(4 - f(x))^2 - (4 - g(x))^2] dx$$

$$\pi \int_0^2 \left[\left(4 - (2x^2 - 6x + 4) \right)^2 - \left(4 - 4\cos\left(\frac{\pi}{4}x\right) \right)^2 \right] dx$$

8-11-12

52. Let R be the region enclosed by the graph of $y = \frac{x^2}{x^2+1}$, the line $x=1$, and the x -axis.

Find the volume of the solid generated when R is rotated about the y -axis.

Part B

1 point is earned for setup

2 point are earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned}
 \text{Volume} &= 2\pi \int_0^1 x \left(\frac{x^2}{x^2+1} \right) dx \\
 &= 2\pi \int_0^1 x - \frac{x}{x^2+1} dx \\
 &= 2\pi \left(\frac{x^2}{2} - \frac{1}{2} \ln |x^2+1| \right) \Big|_0^1 \\
 &= \pi(1 - \ln 2)
 \end{aligned}$$

or

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^{1/2} \left(1 - \frac{y}{1-y} \right) dy \\
 &= \pi(2y + \ln |y-1|) \Big|_0^{1/2} \\
 &= \pi(1 - \ln 2)
 \end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all four of the following points:

1 point is earned for setup

2 point are earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned}
 \text{Volume} &= 2\pi \int_0^1 x \left(\frac{x^2}{x^2+1} \right) dx \\
 &= 2\pi \int_0^1 x - \frac{x}{x^2+1} dx \\
 &= 2\pi \left(\frac{x^2}{2} - \frac{1}{2} \ln |x^2+1| \right) \Big|_0^1 \\
 &= \pi(1 - \ln 2)
 \end{aligned}$$

or

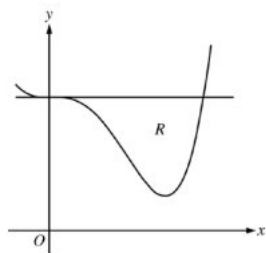
8-11-12

$$\text{Volume} = \pi \int_0^{1/2} \left(1 - \frac{y}{1-y}\right) dy$$

$$= \pi(2y + \ln|y-1|) \Big|_0^{1/2}$$

$$= \pi(1 - \ln 2)$$

53.



Let R be the region enclosed by the graph of $f(x) = x^4 - 2.3x^3 + 4$ and the horizontal line $y = 4$, as shown in the figure above.

Find the volume of the solid generated when R is rotated about the horizontal line $y = -2$.

Part A

2 points are earned for the integrand

1 point is earned for limits

1 point is earned for answer

$$f(x) = 4 \Rightarrow x = 0, 2.3$$

$$\text{Volume} = \pi \int_0^{2.3} [(4+2) - (f(x)+2)] dx$$

$$= 98.868 \text{ (or } 98.867)$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for limits

1 point is earned for answer

$$f(x) = 4 \Rightarrow x = 0, 2.3$$

$$\text{Volume} = \pi \int_0^{2.3} [(4+2) - (f(x)+2)] dx$$

$$= 98.868 \text{ (or } 98.867)$$

8-11-12

54. Let R be the region in the first and second quadrants bounded above by the graph of $y = \frac{20}{1+x^2}$ and below by the horizontal line $y = 2$.



Find the volume of the solid generated when R is rotated about the x -axis.

Part B

2 points are earned for the integrand

1 point is earned for the answer

$$Volume = \pi \int_{-3}^3 \left(\left(\frac{20}{1+x^2} \right)^2 - 2^2 \right) dx = 1871.190$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

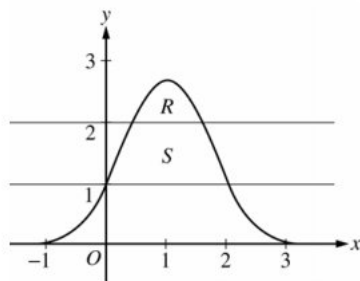
2 points are earned for the integrand

1 point is earned for the answer

$$Volume = \pi \int_{-3}^3 \left(\left(\frac{20}{1+x^2} \right)^2 - 2^2 \right) dx = 1871.190$$

8-11-12

55.



Let R be the region bounded by the graph of $y = e^{2x-x^2}$ and the horizontal line $y = 2$, and let S be the region bounded by the graph of $y = e^{2x-x^2}$ and the horizontal lines $y = 1$ and $y = 2$, as shown above.

Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 1$.

Part C

2 points are earned for the integrand

1 point is earned for the constant and limits

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Volume} = \pi \int_P^Q ((e^{2x-x^2} - 1)^2 - (2 - 1)^2) dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the constant and limits

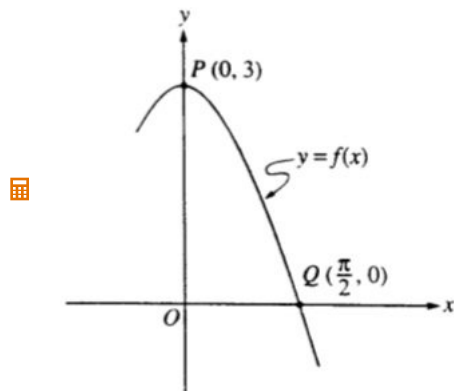
$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Volume} = \pi \int_P^Q ((e^{2x-x^2} - 1)^2 - (2 - 1)^2) dx$$

8-11-12

56.



Let f be the function given by $f(x) = 3 \cos x$. As shown above, the graph of f crosses the y -axis at point P and the x -axis at point Q .

- Write an equation for the line passing through the points P and Q .
- Write an equation for the line tangent to the graph of f at point Q . Show the analysis that leads to your conclusion.
- Find the x -coordinate of the point on the graph of f , between points P and Q , at which the line tangent to the graph of f is parallel to line PQ .
- Let R be the region in the first quadrant bounded by the graph of f and line segment PQ . Write an integral expression for the volume of the solid generated by revolving the region R about the x -axis. Do not evaluate.

Part A

1 point is earned for the correct slope

$$\text{slope} = \frac{3-0}{0-\pi/2} = -\frac{6}{\pi}$$

1 point is earned for the correct equation

$$y - 3 = -\frac{6}{\pi}(x - 0)$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct slope

$$\text{slope} = \frac{3-0}{0-\pi/2} = -\frac{6}{\pi}$$

1 point is earned for the correct equation

$$y - 3 = -\frac{6}{\pi}(x - 0)$$

Part B

1 point is earned for the correct statement $f'(\pi/2) = -3$

1 point is earned for an equation using $f'(\pi/2)$ (or using a derivation) and $f(\pi/2)$

1/2 if equations only

$$f'(x) = -3 \sin x$$

$$f'(\pi/2) = -3 \sin(\pi/2) = -3$$

$$y - 0 = -3(x - \pi/2)$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the correct statement $f'(\pi/2) = -3$

8-11-12

1 point is earned for an equation using $f'(\pi/2)$ (or using a derivation) and $f(\pi/2)$

1/2 if equations only

$$f'(x) = -3 \sin x$$

$$f'(\pi/2) = -3 \sin(\pi/2) = -3$$

$$y - 0 = -3(x - \pi/2)$$

Part C

1 point is earned for correctly equating the derivative to the slope

$$f'(x) = -3 \sin x = -\frac{6}{\pi}$$

$$\sin x = \frac{2}{\pi}$$

1 point is earned for the correct solution in $[0, \frac{\pi}{2}]$

1/2 if solution only

no credit for "no solution exists"

$$x = 0.690$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for correctly equating the derivative to the slope

$$f'(x) = -3 \sin x = -\frac{6}{\pi}$$

$$\sin x = \frac{2}{\pi}$$

1 point is earned for the correct solution in $[0, \frac{\pi}{2}]$

1/2 if solution only

no credit for "no solution exists"

$$x = 0.690$$

Part D

2 points are earned for the correct integrand

0/2 if not difference of 2 squares

1/2 if incorrect but of form $(a \cos x)^2 - (bx + c)^2$; $a, b \neq 0$

1/2 if reversed

1 point is earned for the correct constant and limits

0/3 for shell around y-axis

$$V = \pi \int_0^{\pi/2} \left[(3 \cos x)^2 - \left(-\frac{6}{\pi}x + 3 \right)^2 \right] dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the correct integrand

0/2 if not difference of 2 squares

1/2 if incorrect but of form $(a \cos x)^2 - (bx + c)^2$; $a, b \neq 0$

8-11-12

1/2 if reversed

1 point is earned for the correct constant and limits

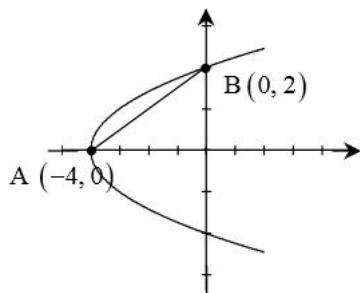
0/3 for shell around y -axis

$$V = \pi \int_0^{\pi/2} \left[(3 \cos x)^2 - \left(-\frac{6}{\pi}x + 3 \right)^2 \right] dx$$

8-11-12

57. Consider the curve $y^2 = 4 + x$ and chord AB joining the points $A(-4, 0)$ and $B(0, 2)$ on the curve.
- Find the x - and y -coordinates of the point on the curve where the tangent line is parallel to chord AB .
 - Find the area of the region R enclosed by the curve and the chord AB .
 - Find the volume of the solid generated when the region R , defined in part (b), is revolved about the x -axis.

Part A



$$\text{slope of } AB = \frac{2}{4} = \frac{1}{2}$$

Method 1:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$y = \sqrt{4 + x}; \quad \frac{dy}{dx} = \frac{1}{2\sqrt{4 + x}}; \quad \frac{1}{2\sqrt{4 + x}} = \frac{1}{2}$$

1 point is earned for the correct answer

$$\text{so } x = -3, y = 1$$

Method 2:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$2y \frac{dy}{dx} = 1, \text{ so } 2y \left(\frac{1}{2} \right) = 1$$

1 point is earned for the correct answer

$$\text{and } y = 1, x = -3$$

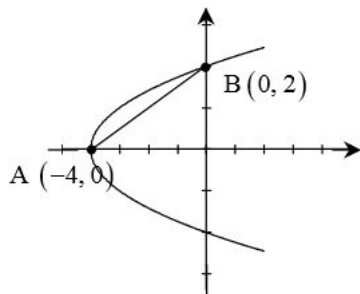
Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

8-11-12



$$\text{slope of } AB = \frac{2}{4} = \frac{1}{2}$$

Method 1:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$y = \sqrt{4+x}; \frac{dy}{dx} = \frac{1}{2\sqrt{4+x}}; \frac{1}{2\sqrt{4+x}} = \frac{1}{2}$$

1 point is earned for the correct answer

$$\text{so } x = -3, y = 1$$

Method 2:

1 point is earned for the correct differentiation

1 point is earned for correctly setting student's derivative = student's slope of AB

$$2y \frac{dy}{dx} = 1, \text{ so } 2y \left(\frac{1}{2} \right) = 1$$

1 point is earned for the correct answer

$$\text{and } y = 1, x = -3$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

Part BMethod 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\begin{aligned} \int_{-4}^0 \left[\sqrt{4+x} - \left(\frac{1}{2}x + 2 \right) \right] dx &= \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0 \\ &= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3} \end{aligned}$$

Method 2:

8-11-12

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_0^2 [(2y - 4) - (y^2 - 4)] dy = y^2 - \frac{y^3}{3} \Big|_0^2$$

$$= 4 - \frac{8}{3} = \frac{4}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_{-4}^0 \left[(\sqrt{4+x}) - \left(\frac{1}{2}x + 2 \right) \right] dx = \frac{2}{3}(4+x)^{3/2} - \frac{1}{4}x^2 - 2x \Big|_{-4}^0$$

$$= \frac{2}{3}(4)^{3/2} - (-4 + 8) = \frac{16}{3} - 4 = \frac{4}{3}$$

Method 2:

8-11-12

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_0^2 [(2y - 4) - (y^2 - 4)] dy = y^2 - \frac{y^3}{3} \Big|_0^2$$

$$= 4 - \frac{8}{3} = \frac{4}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of areas

1 point is earned for the correct answer

$$\int_{-4}^0 \sqrt{4+x} dx = \frac{16}{3}; \text{ Area of triangle} = 4$$

$$\text{Area of region} = \frac{16}{3} - 4 = \frac{4}{3}$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

Part C

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

8-11-12

$$\begin{aligned} & \pi \int_{-4}^0 \left((\sqrt{4+x})^2 - \left(\frac{1}{2}x + 2 \right)^2 \right) dx \\ &= \pi \int_{-4}^0 \left(4 + x - \left(\frac{1}{4}x^2 + 2x + 4 \right) \right) dx \\ &= \pi(8 - 16/3) = 8\pi/3 \approx 8.378 \end{aligned}$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\int_0^2 2\pi y [(2y - 4) - (y^2 - 4)] dy = \frac{8\pi}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\pi \int_{-4}^0 (\sqrt{4+x})^2 dx = \pi \left(4x + \frac{x^2}{2} \right) \bigg|_{-4}^0$$

$$= 0 - \pi(-16 + 8) = 8\pi$$

$$\text{Volume of cone} = 1/3 \pi (2)^2 (4) = 16\pi/3$$

$$\text{Volume} = 8\pi - 16\pi/3 = 8\pi/3$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.



0	1	2	3
---	---	---	---

8-11-12

The student response earns all of the following points:

Method 1:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\begin{aligned} \pi \int_{-4}^0 \left((\sqrt{4+x})^2 - \left(\frac{1}{2}x + 2 \right)^2 \right) dx \\ = \pi \int_{-4}^0 \left(4+x - \left(\frac{1}{4}x^2 + 2x + 4 \right) \right) dx \end{aligned}$$

$$= \pi(8 - 16/3) = 8\pi/3 \approx 8.378$$

Method 2:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\int_0^2 2\pi y [(2y-4) - (y^2-4)] dy = \frac{8\pi}{3}$$

Method 3:

1 point is earned for the correct bounds and constant

1 point is earned for the correct integrand

1 point is earned for the correct antidifferentiation and evaluation

Note: 0 of 1 if integral not of the form $\int_a^b ([f(x)]^2 - [g(x)]^2) dx$ or $\int_c^d y[f(y) - g(y)] dy$

OR

1 point is earned for the correct bounds and constant

8-11-12

1 point is earned for the correct subtraction of volumes

1 point is earned for the correct answer

$$\pi \int_{-4}^0 (\sqrt{4+x})^2 dx = \pi \left(4x + \frac{x^2}{2} \right) \bigg|_{-4}^0$$

$$= 0 - \pi(-16 + 8) = 8\pi$$

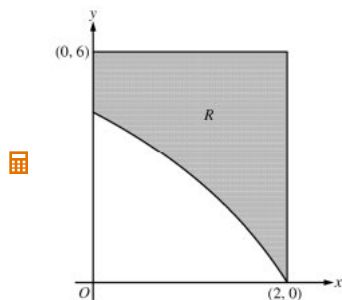
$$\text{Volume of cone} = \frac{1}{3} \pi (2)^2 (4) = 16\pi/3$$

$$\text{Volume} = 8\pi - 16\pi/3 = 8\pi/3$$

Note: In (a), (b), and (c) any arithmetic error results in loss of last point.

8-11-12

58.



In the figure above, R is the shaded region in the first quadrant bounded by the graph of $y = 4\ln(3 - x)$, the horizontal line $y = 6$, and the vertical line $x = 2$.

- (a) Find the area of R .
 (b) Find the volume of the solid generated when R is revolved about the horizontal line $y = 8$.
 (c) The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of the solid.

Part A

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3 - x)) dx = 6.816 \text{ or } 6.817$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3 - x)) dx = 6.816 \text{ or } 6.817$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

Part B

2 points are earned for the integrand

1 point is earned for the answer

$$\pi \int_0^2 ((8 - 4\ln(3 - x))^2 - (8 - 6)^2) dx$$

$$= 168.179 \text{ or } 168.180$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

8-11-12

1 point is earned for the answer

$$\pi \int_0^2 \left((8 - 4\ln(3 - x))^2 - (8 - 6)^2 \right) dx$$

$$= 168.179 \text{ or } 168.180$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

Part C

2 points are earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3 - x))^2 dx = 26.266 \text{ or } 26.267$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

✓				
0	1	2	3	4

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\int_0^2 (6 - 4\ln(3 - x))^2 dx = 26.266 \text{ or } 26.267$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

59. Let R be the region in the first quadrant bounded by the graph of $y = \sqrt{x-1}$, the x -axis, and the vertical line $x = 10$. Which of the following integrals gives the volume of the solid generated by revolving R about the y -axis?

(A) $\pi \int_1^{10} (x-1) \, dx$

(B) $\pi \int_1^{10} (100 - (x-1)) \, dx$

(C) $\pi \int_0^3 \left(10 - (y^2 + 1) \right)^2 \, dy$

(D) $\pi \int_0^3 \left(100 - (y^2 + 1) \right)^2 \, dy$ ✓

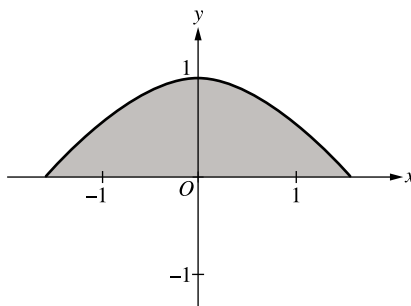
Answer D

Correct. Rotating a typical slice about the y -axis will form a washer with inner radius r_1 and outer radius r_2 , where each radius must be expressed in terms of y . Since $y = \sqrt{x-1}$, the inner radius is $r_1 = x = y^2 + 1$. Since the region is bounded on the right by the vertical line $x = 10$, the outer radius is $r_2 = 10$. The cross-sectional area of a typical slice rotated about the y -axis is therefore $A(y) = \pi r_2^2 - \pi r_1^2 = \pi (10^2 - (y^2 + 1)^2)$. The volume of the solid is found by using the definite integral of the cross-sectional area for y between 0 and $\sqrt{10-1} = 3$, as follows.

$$\int_0^3 \pi (10^2 - (y^2 + 1)^2) dy = \pi \int_0^3 (100 - (y^2 + 1)^2) dy$$

8-11-12

60.



Let R be the shaded region enclosed by the graph of $y = \cos x$ and the x -axis for $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$, as shown in the figure. Which of the following gives the volume of the solid generated when R is revolved about the horizontal line $y = -1$?

(A) $\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} ((\cos x + 1) - 1)^2 dx$

(B) $\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} ((\cos x)^2 - 1) dx$

(C) $\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} ((\cos x - 1)^2 - 1) dx$

(D) $\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} ((\cos x + 1)^2 - 1) dx$ ✓

Answer D

Correct. For a solid of revolution with a horizontal axis, $V = \pi \int_a^b ((R(x))^2 - (r(x))^2) dx$ where for this region, $R(x)$ is the distance from the graph to the horizontal line $y = -1$, and $r(x)$ is the distance from the x -axis to the horizontal line $y = -1$. Therefore, $R(x) = \cos x - (-1) = \cos x + 1$ and $r(x) = 1$.

8-11-12

61. Let R be the region between the graphs of $y = 1 + \sin(\pi x)$ and $y = x^2$ from $x = 0$ to $x = 1$.

(a) Find the area of R .

(b) Set up, but do not integrate an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the x -axis.

(c) Set up, but do not integrate an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part A

2 points are earned for integral

1 point is **deducted** for limit error

2 points are **deducted** for integrand error

2 points are **deducted** for reversed integral

3 points are earned for antiderivative and evaluate

3 points are **deducted** for constant term

2 points are **deducted** for chain rule error

1 point is **deducted** for each other error

$$\begin{aligned} A &= \int_0^1 1 + \sin \pi x - x^2 dx \\ &= \left(x - \frac{1}{\pi} \cos \pi x - \frac{1}{3} x^3 \right) \Big|_0^1 \\ &= \left(1 - \frac{1}{\pi}(-1) - \frac{1}{3} \right) - \left(0 - \frac{1}{\pi} - 0 \right) \\ &= \frac{2}{3} + \frac{2}{\pi} \end{aligned}$$

✓

0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

2 points are earned for integral

1 point is **deducted** for limit error

2 points are **deducted** for integrand error

2 points are **deducted** for reversed integral

3 points are earned for antiderivative and evaluate

3 points are **deducted** for constant term

2 points are **deducted** for chain rule error

1 point is **deducted** for each other error

$$\begin{aligned} A &= \int_0^1 1 + \sin \pi x - x^2 dx \\ &= \left(x - \frac{1}{\pi} \cos \pi x - \frac{1}{3} x^3 \right) \Big|_0^1 \\ &= \left(1 - \frac{1}{\pi}(-1) - \frac{1}{3} \right) - \left(0 - \frac{1}{\pi} - 0 \right) \\ &= \frac{2}{3} + \frac{2}{\pi} \end{aligned}$$

Part B

2 points are earned for integral expression

1 point is **deducted** for incorrect limits

8-11-12

$$V = \pi \int_0^1 (1 + \sin \pi x)^2 - x^4 dx$$

or

$$2\pi \int_0^1 y^{3/2} dy + 2\pi \int_1^2 y \left(1 - \frac{2}{\pi} \arcsin(y-1)\right) dy$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for integral expression

1 point is **deducted** for incorrect limits

$$V = \pi \int_0^1 (1 + \sin \pi x)^2 - x^4 dx$$

or

$$2\pi \int_0^1 y^{3/2} dy + 2\pi \int_1^2 y \left(1 - \frac{2}{\pi} \arcsin(y-1)\right) dy$$

Part C

2 points are earned for integral

1 point is **deducted** for π or 2π error2 points are **deducted** for integrand error (including reverse)1 point is **deducted** for one or more limit error

$$V = 2\pi \int_0^1 x(1 + \sin(\pi x) - x^2) dx$$

or

$$\pi \int_0^1 y dy + \pi \int_1^2 \left(1 - \frac{1}{\pi} \arcsin(y-1)\right)^2 - \left(\frac{1}{\pi} \arcsin(y-1)\right)^2 dy$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for integral

1 point is **deducted** for π or 2π error2 points are **deducted** for integrand error (including reverse)1 point is **deducted** for one or more limit error

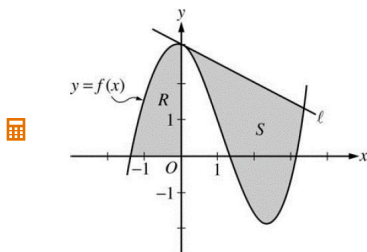
$$V = 2\pi \int_0^1 x(1 + \sin(\pi x) - x^2) dx$$

or

$$\pi \int_0^1 y dy + \pi \int_1^2 \left(1 - \frac{1}{\pi} \arcsin(y-1)\right)^2 - \left(\frac{1}{\pi} \arcsin(y-1)\right)^2 dy$$

8-11-12

62.



Let f be the function given by $f(x) = \frac{x^3}{4} - \frac{x^2}{3} - \frac{x}{2} + 3 \cos x$. Let R be the shaded region in the second quadrant bounded by the graph of f , and let S be the shaded region bounded by the graph of f and line ℓ , the line tangent to the graph of f at $x = 0$, as shown above.

- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is rotated about the horizontal line $y = -2$.
- (c) Write, but do not evaluate, an integral expression that can be used to find the area of S .

Part A

1 point is earned for integral

1 point is earned for the answer

For $x < 0$, $f(x) = 0$ when $x = -1.37312$

Let $P = -1.37312$.

$$\text{Area of } R = \int_P^0 f(x) dx = 2.903$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integral

1 point is earned for the answer

For $x < 0$, $f(x) = 0$ when $x = -1.37312$

Let $P = -1.37312$.

$$\text{Area of } R = \int_P^0 f(x) dx = 2.903$$

Part B

1 point is earned for limits and constant

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_P^0 \left((f(x) + 2)^2 - 4 \right) dx = 59.361$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limits and constant

8-11-12

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_P^0 \left((f(x) + 2)^2 - 4 \right) dx = 59.361$$

Part C

1 point is earned for the tangent line

1 point is earned for the integrand

1 point is earned for the limits

The equation of the tangent line ℓ is $y = 3 - \frac{1}{2}x$.

The graph of f and line ℓ intersect at $A = 3.38987$.

$$\text{Area of } S = \int_0^A \left(\left(3 - \frac{1}{2}x \right) - f(x) \right) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the tangent line

1 point is earned for the integrand

1 point is earned for the limits

The equation of the tangent line ℓ is $y = 3 - \frac{1}{2}x$.

The graph of f and line ℓ intersect at $A = 3.38987$.

$$\text{Area of } S = \int_0^A \left(\left(3 - \frac{1}{2}x \right) - f(x) \right) dx$$

8-11-12

63. Let R be the region enclosed by the graphs of $y = e^x$, $y = (x - 1)^2$, and the line $x = 1$.

(a) Find the area of R .

(b) Find the volume of the solid generated when R is revolved about the x -axis.

(c) Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part A

1 point is earned for definite integral

1 point is **deducted** for incorrect limits

1 point is **deducted** for reversal

1 point is earned for anti derivative

1 point is earned for evaluation

$$\begin{aligned} A &= \int_0^1 e^x - (x - 1)^2 dx \\ &= \int_0^1 e^x - x^2 + 2x - 1 dx \\ &= e^x \Big|_0^1 - \frac{1}{3}(x - 1)^3 \Big|_0^1 \\ &= (e - 1) - \frac{1}{3} = e - \frac{4}{3} \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for definite integral

1 point is **deducted** for incorrect limits

1 point is **deducted** for reversal

1 point is earned for anti derivative

1 point is earned for evaluation

$$\begin{aligned} A &= \int_0^1 e^x - (x - 1)^2 dx \\ &= \int_0^1 e^x - x^2 + 2x - 1 dx \\ &= e^x \Big|_0^1 - \frac{1}{3}(x - 1)^3 \Big|_0^1 \\ &= (e - 1) - \frac{1}{3} = e - \frac{4}{3} \end{aligned}$$

Part B

2 points are earned for definite integral

2 points are **deducted** for faulty f , g , or u

1 point is earned for correct antiderivative

1 point is earned for evaluation of anti derivative

8-11-12

$$\begin{aligned}
 V &= \pi \int_0^1 e^{2x} - (x-1)^4 dx \\
 &= \pi \left[\frac{e^{2x}}{2} \right]_0^1 - \pi \left[\frac{1}{5}(x-1)^5 \right]_0^1 \\
 &= \pi \left[\left(\frac{e^2}{2} - \frac{1}{2} \right) - \frac{1}{5} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right)
 \end{aligned}$$

or

$$\begin{aligned}
 V &= 2\pi \int_0^1 y [1 - (1 - \sqrt{y})] dy + 2\pi \int_1^e y (1 - \ln y) dy \\
 &= 2\pi \cdot \frac{2}{5} y^{5/2} \Big|_0^1 + 2\pi \left[\frac{1}{2} y^2 - \left(\frac{1}{2} y^2 \ln y - \frac{1}{4} y^2 \right) \right]_1^e \\
 &= \frac{4}{5} \pi + 2\pi \left[\frac{1}{4} e^2 - \frac{3}{4} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right)
 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for definite integral

2 points are **deducted** for faulty f, g, or u

1 point is earned for correct antiderivative

1 point is earned for evaluation of anti derivative

$$\begin{aligned}
 V &= \pi \int_0^1 e^{2x} - (x-1)^4 dx \\
 &= \pi \left[\frac{e^{2x}}{2} \right]_0^1 - \pi \left[\frac{1}{5}(x-1)^5 \right]_0^1 \\
 &= \pi \left[\left(\frac{e^2}{2} - \frac{1}{2} \right) - \frac{1}{5} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right)
 \end{aligned}$$

or

$$\begin{aligned}
 V &= 2\pi \int_0^1 y [1 - (1 - \sqrt{y})] dy + 2\pi \int_1^e y (1 - \ln y) dy \\
 &= 2\pi \cdot \frac{2}{5} y^{5/2} \Big|_0^1 + 2\pi \left[\frac{1}{2} y^2 - \left(\frac{1}{2} y^2 \ln y - \frac{1}{4} y^2 \right) \right]_1^e \\
 &= \frac{4}{5} \pi + 2\pi \left[\frac{1}{4} e^2 - \frac{3}{4} \right] = \pi \left(\frac{e^2}{2} - \frac{7}{10} \right)
 \end{aligned}$$

Part C

2 points are earned for Definite integral

2 points are **deducted** for f, g, or u error1 point is **deducted** for limit error, K error and integrand reversal

$$V = 2\pi \int_0^1 x [e^x - (x-1)^2] dx$$

or

$$V = \pi \int_0^1 1 - (1 - \sqrt{y})^2 dy + \pi \int_1^e 1 - (\ln y)^2 dy$$

8-11-12

✓

0	1	2
---	---	---

The student response earns all of the following points:

2 points are earned for Definite integral

2 points are **deducted** for f, g, or u error

1 point is **deducted** for limit error, K error and integrand reversal

$$V = 2\pi \int_0^1 x \left[e^x - (x-1)^2 \right] dx$$

or

$$V = \pi \int_0^1 1 - (1 - \sqrt{y})^2 dy + \pi \int_1^e 1 - (\ln y)^2 dy$$

8-11-12

64.  Let R be the region in the first and second quadrants bounded above by the graph of $y = \frac{20}{1+x^2}$ and below by the horizontal line $y = 2$.
- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is rotated about the x -axis.
- (c) The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are semicircles. Find the volume of this solid.

Part A

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right) dx = 37.961 \text{ or } 37.962$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right) dx = 37.961 \text{ or } 37.962$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

Part B

2 points are earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_{-3}^3 \left(\left(\frac{20}{1+x^2} \right)^2 - 2^2 \right) dx = 1871.190$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

8-11-12

$$\text{Volume} = \pi \int_{-3}^3 \left(\left(\frac{20}{1+x^2} \right)^2 - 2^2 \right) dx = 1871.190$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

Part C

2 points are earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \frac{\pi}{2} \int_{-3}^3 \left(\frac{1}{2} \left(\frac{20}{1+x^2} - 2 \right) \right)^2 dx \\ &= \frac{\pi}{8} \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right)^2 dx = 174.268 \end{aligned}$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \frac{\pi}{2} \int_{-3}^3 \left(\frac{1}{2} \left(\frac{20}{1+x^2} - 2 \right) \right)^2 dx \\ &= \frac{\pi}{8} \int_{-3}^3 \left(\frac{20}{1+x^2} - 2 \right)^2 dx = 174.268 \end{aligned}$$

1 point is earned for the correct limits in an integral in (a), (b), or (c)

$$\frac{20}{1+x^2} = 2 \text{ when } x = \pm 3$$

8-11-12

65. Let R be the region enclosed by the graph of $y = x^2 / x^2 + 1$, the line $x = 1$, and the x -axis.

(a) Find the area of R .

(b) Find the volume of the solid generated when R is rotated about the y -axis.

Part A

1 point is earned for bounds

1 point is earned for integrand

2 points are earned for antiderivative

2 points **deducted** for trig calculation error

1 point is earned for evaluation

$$\begin{aligned}\text{Area} &= \int_0^1 \frac{x^2}{x^2 + 1} dx \\ &= \int_0^1 1 - \frac{1}{x^2 + 1} dx \\ &= x - \arctan x \Big|_0^1 \\ &= 1 - \frac{\pi}{4}\end{aligned}$$

✓

0	1	2	3	4	5
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The student response earns all of the following points:

1 point is earned for bounds

1 point is earned for integrand

2 points are earned for antiderivative

2 points **deducted** for trig calculation error

1 point is earned for evaluation

$$\begin{aligned}\text{Area} &= \int_0^1 \frac{x^2}{x^2 + 1} dx \\ &= \int_0^1 1 - \frac{1}{x^2 + 1} dx \\ &= x - \arctan x \Big|_0^1 \\ &= 1 - \frac{\pi}{4}\end{aligned}$$

Part B

1 point is earned for setup

2 point are earned for antiderivative

1 point is earned for evaluation

$$\begin{aligned}\text{Volume} &= 2\pi \int_0^1 x \left(\frac{x^2}{x^2 + 1} \right) dx \\ &= 2\pi \int_0^1 x - \frac{x}{x^2 + 1} dx \\ &= 2\pi \left(\frac{x^2}{2} - \frac{1}{2} \ln |x^2 + 1| \right) \Big|_0^1 \\ &= \pi(1 - \ln 2)\end{aligned}$$

or

8-11-12

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^{\frac{1}{2}} x \left(1 - \frac{y}{1-y} \right) dy \\
 &= \pi (2y + \ln |y-1|) \Big|_0^{\frac{1}{2}} \\
 &= \pi (1 - \ln 2)
 \end{aligned}$$

✓

0	1	2	3	4
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The student response earns all of the following points:

1 point is earned for setup

2 point are earned for antiderivative

1 point is earned for evaluation

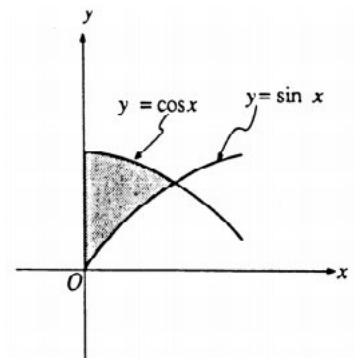
$$\begin{aligned}
 \text{Volume} &= 2\pi \int_0^1 x \left(\frac{x^2}{x^2+1} \right) dx \\
 &= 2\pi \int_0^1 x - \frac{x}{x^2+1} dx \\
 &= 2\pi \left(\frac{x^2}{2} - \frac{1}{2} \ln |x^2+1| \right) \Big|_0^1 \\
 &= \pi (1 - \ln 2)
 \end{aligned}$$

or

$$\begin{aligned}
 \text{Volume} &= \pi \int_0^{\frac{1}{2}} x \left(1 - \frac{y}{1-y} \right) dy \\
 &= \pi (2y + \ln |y-1|) \Big|_0^{\frac{1}{2}} \\
 &= \pi (1 - \ln 2)
 \end{aligned}$$

8-11-12

66.



Let R be the shaded region in the first quadrant enclosed by the y -axis and the graphs of $y = \sin x$ and $y = \cos x$, as shown in the figure above.

- Find the area of R .
- Find the volume of the solid generated when R is revolved about the x -axis.
- Find the volume of the solid whose base is R and whose cross sections cut by planes perpendicular to the x -axis are squares.

Part A

1 point is earned for limits

1 point is earned for integrand

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 \text{Area} &= \int_0^{\pi/4} \cos x - \sin x \, dx \\
 &= (\sin x + \cos x) \Big|_0^{\pi/4} \\
 &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) \\
 &= \sqrt{2} - 1
 \end{aligned}$$

✓

0	1	2	3
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The student response earns all of the following points:

1 point is earned for limits

1 point is earned for integrand

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 \text{Area} &= \int_0^{\pi/4} \cos x - \sin x \, dx \\
 &= (\sin x + \cos x) \Big|_0^{\pi/4} \\
 &= \left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} \right) - (0 + 1) \\
 &= \sqrt{2} - 1
 \end{aligned}$$

Part B

1 point is earned for limits

1 point is earned for integrand

1 point is **deducted** for constant error

1 point is earned for antiderivative and evaluation

8-11-12

$$\begin{aligned}
 V &= \pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx \\
 &= \pi \int_0^{\pi/4} \cos 2x \, dx \\
 &= \frac{\pi}{2} \sin 2x \Big|_0^{\pi/4} \\
 &= \frac{\pi}{2} (1 - 0) = \frac{\pi}{2}
 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for limits

1 point is earned for integrand

1 point is **deducted** for constant error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \pi \int_0^{\pi/4} \cos^2 x - \sin^2 x \, dx \\
 &= \pi \int_0^{\pi/4} \cos 2x \, dx \\
 &= \frac{\pi}{2} \sin 2x \Big|_0^{\pi/4} \\
 &= \frac{\pi}{2} (1 - 0) = \frac{\pi}{2}
 \end{aligned}$$

Part C

2 points are earned for integral

1 point is **deducted** for constant or limit error

1 point is earned for antiderivative and evaluation

$$\begin{aligned}
 V &= \int_0^{\pi/4} (\cos x - \sin x)^2 \, dx \\
 &= \int_0^{\pi/4} 1 - 2 \sin x \cos x \, dx \\
 &= (x - \sin^2 x) \Big|_0^{\pi/4} \\
 &= \frac{\pi}{4} - \frac{1}{2} - (0 - 0) \\
 &= \frac{\pi}{4} - \frac{1}{2}
 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integral

1 point is **deducted** for constant or limit error

1 point is earned for antiderivative and evaluation

8-11-12

$$\begin{aligned} V &= \int_0^{\pi/4} (\cos x - \sin x)^2 dx \\ &= \int_0^{\pi/4} 1 - 2 \sin x \cos x \, dx \\ &= \left(x - \sin^2 x \right) \Big|_0^{\pi/4} \\ &= \frac{\pi}{4} - \frac{1}{2} - (0 - 0) \\ &= \frac{\pi}{4} - \frac{1}{2} \end{aligned}$$

8-11-12

67. Let R be the region in the first quadrant enclosed by the graph of $y = \sqrt{6x + 4}$, the line $y = 2x$, and the y -axis.

(a) Find the area of R .

(b) Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the x -axis.

(c) Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated when R is revolved about the y -axis.

Part A

1 point is earned for limit

2 point are earned for integrand

1 point is earned for antiderivative

1 point is earned for $x = 3$

$$\begin{aligned}\text{Area} &= \int_0^2 \sqrt{6x + 4} - 2x \, dx \\ &= \frac{1}{6} \cdot \frac{2}{3} (6x + 4)^{3/2} - x^2 \Big|_0^2 \\ &= \left(\frac{64}{9} - 4 \right) - \frac{8}{9} = \boxed{\frac{20}{9}}\end{aligned}$$

✓

0	1	2	3	4	5
---	---	---	---	---	---

The student response earns all of the following points:

1 point is earned for limit

2 point are earned for integrand

1 point is earned for antiderivative

1 point is earned for $x = 3$

$$\begin{aligned}\text{Area} &= \int_0^2 \sqrt{6x + 4} - 2x \, dx \\ &= \frac{1}{6} \cdot \frac{2}{3} (6x + 4)^{3/2} - x^2 \Big|_0^2 \\ &= \left(\frac{64}{9} - 4 \right) - \frac{8}{9} = \boxed{\frac{20}{9}}\end{aligned}$$

Part B

1 point is earned for integrand

1 point is earned for limits and constant

2 points **deducted** for incorrect integrand

1 point **deducted** for wrong limits, wrong constant or reversal

Volume about x -axis

$$\begin{aligned}\text{Disks} &= \pi \int_0^2 (6x + 4) - 4x^2 \, dx \\ &\quad \text{or} \\ &= \pi \int_0^2 (6x + 4) \, dx - \frac{32\pi}{3}\end{aligned}$$

✓

0	1	2
---	---	---

8-11-12

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for limits and constant

1 point **deducted** for wrong limits, wrong constant or reversal

Volume about x -axis

$$\begin{aligned}\text{Disks} &= \pi \int_0^2 (6x + 4) - 4x^2 \, dx \\ &\quad \text{or} \\ &= \pi \int_0^2 (6x + 4) \, dx - \frac{32\pi}{3}\end{aligned}$$

Part C

1 point is earned for integrand

1 point is earned for limits and constant

1 point **deducted** for wrong limits, wrong constant or reversal

Volume about y -axis

$$\begin{aligned}\text{Shells} &= 2\pi \int_0^2 x \left(\sqrt{6x + 4} - 2x \right) \, dx \\ \text{Disks} &= \pi \int_0^4 \left(\frac{y}{2} \right)^2 \, dy - \pi \int_2^4 \left(\frac{y^2 - 4}{6} \right)^2 \, dy\end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for limits and constant

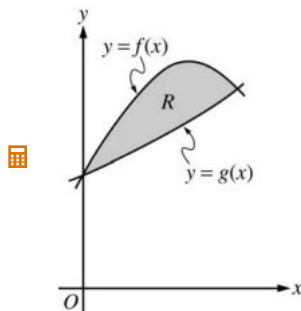
1 point **deducted** for wrong limits, wrong constant or reversal

Volume about y -axis

$$\begin{aligned}\text{Shells} &= 2\pi \int_0^2 x \left(\sqrt{6x + 4} - 2x \right) \, dx \\ \text{Disks} &= \pi \int_0^4 \left(\frac{y}{2} \right)^2 \, dy - \pi \int_2^4 \left(\frac{y^2 - 4}{6} \right)^2 \, dy\end{aligned}$$

8-11-12

68.



Let f and g be the functions given by $f(x) = 1 + \sin(2x)$ and $g(x) = e^{x/2}$. Let R be the shaded region in the first quadrant enclosed by the graphs of f and g as shown in the figure above.

(a) Find the area of R .

(b) Find the volume of the solid generated when R is revolved about the x -axis.

(c) The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are semicircles with diameters extending from $y = f(x)$ to $y = g(x)$. Find the volume of this solid.

Part A

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Area} &= \int_0^S (f(x) - g(x))dx \\ &= \int_0^S (1 + \sin(2x) - e^{x/2})dx \\ &= 0.429\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

✓			
0	1	2	3

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Area} &= \int_0^S (f(x) - g(x))dx \\ &= \int_0^S (1 + \sin(2x) - e^{x/2})dx \\ &= 0.429\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

Part B

2 point is earned for the integrand

<-1> for each error

Note: 0/2 if integral not of form $c \int_a^b (R^2(x) - r^2(x))dx$

8-11-12

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^S \left((f(x))^2 - (g(x))^2 \right) dx \\ &= \pi \int_0^S \left((1 + \sin(2x))^2 - \left(e^{x/2} \right)^2 \right) dx \\ &= 4.266 \text{ or } 4.267\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

✓

0	1	2	3	4
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The student response earns all of the following points:

2 point is earned for the integrand

<-1> for each error

Note: 0/2 if integral not of form $c \int_a^b (R^2(x) - r^2(x)) dx$

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^S \left((f(x))^2 - (g(x))^2 \right) dx \\ &= \pi \int_0^S \left((1 + \sin(2x))^2 - \left(e^{x/2} \right)^2 \right) dx \\ &= 4.266 \text{ or } 4.267\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

Part C

2 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^S \frac{\pi}{2} \left(\frac{f(x) - g(x)}{2} \right)^2 dx \\ &= \int_0^S \frac{\pi}{2} \left(\frac{1 + \sin(2x) - e^{x/2}}{2} \right)^2 dx \\ &= 0.077 \text{ or } 0.078\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 point is earned for the integrand

1 point is earned for the answer

8-11-12

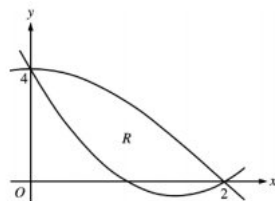
$$\begin{aligned}\text{Volume} &= \int_0^S \frac{\pi}{2} \left(\frac{f(x) - g(x)}{2} \right)^2 dx \\ &= \int_0^S \frac{\pi}{2} \left(\frac{1 + \sin(2x) - e^{x/2}}{2} \right)^2 dx \\ &= 0.077 \text{ or } 0.078\end{aligned}$$

1 point is earned for correct limits in an integral in (a), (b), or (c)

The graphs of f and g intersect in the first quadrant at $(S, T) = (1.13569, 1.76446)$.

8-11-12

69.



Let $f(x) = 2x^2 - 6x + 4$ and $g(x) = 4\cos(1/4 \pi x)$. Let R be the region bounded by the graphs of f and g , as shown in the figure above.

(a) Find the area of R .

(b) Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 4$.

(c) The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Write, but do not evaluate, an integral expression that gives the volume of the solid.

Part A

1 point is earned for the integrand

2 points are earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^2 [g(x) - f(x)] dx \\ &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] dx \\ &= \left[4 \cdot \frac{4}{\pi} \sin\left(\frac{\pi}{4}x\right) - \left(\frac{2x^3}{3} - 3x^2 + 4x\right) \right]_0^2 \\ &= \frac{16}{\pi} - \left(\frac{16}{3} - 12 + 8\right) = \frac{16}{\pi} - \frac{4}{3} \end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

2 points are earned for the antiderivative

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^2 [g(x) - f(x)] dx \\ &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right] dx \\ &= \left[4 \cdot \frac{4}{\pi} \sin\left(\frac{\pi}{4}x\right) - \left(\frac{2x^3}{3} - 3x^2 + 4x\right) \right]_0^2 \\ &= \frac{16}{\pi} - \left(\frac{16}{3} - 12 + 8\right) = \frac{16}{\pi} - \frac{4}{3} \end{aligned}$$

Part B

2 points are earned for the integrand.

1 point is earned for the limits and constant.

$$\begin{aligned} \text{Volume} &= \pi \int_0^2 \left[(4 - f(x))^2 - (4 - g(x))^2 \right] dx \\ &= \pi \int_0^2 \left[(4 - (2x^2 - 6x + 4))^2 - \left(4 - 4 \cos\left(\frac{\pi}{4}x\right)\right)^2 \right] dx \end{aligned}$$

8-11-12

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand.

1 point is earned for the limits and constant.

$$\begin{aligned}\text{Volume} &= \pi \int_0^2 \left[(4 - f(x))^2 - (4 - g(x))^2 \right] dx \\ &= \pi \int_0^2 \left[(4 - (2x^2 - 6x + 4))^2 - \left(4 - 4 \cos\left(\frac{\pi}{4}x\right) \right)^2 \right] dx\end{aligned}$$

Part C

1 point is earned for the integrand

1 point is earned for the limits and constant

$$\begin{aligned}\text{Volume} &= \int_0^2 [g(x) - f(x)]^2 dx \\ &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right]^2 dx\end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

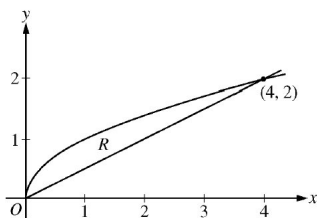
1 point is earned for the integrand

1 point is earned for the limits and constant

$$\begin{aligned}\text{Volume} &= \int_0^2 [g(x) - f(x)]^2 dx \\ &= \int_0^2 \left[4 \cos\left(\frac{\pi}{4}x\right) - (2x^2 - 6x + 4) \right]^2 dx\end{aligned}$$

8-11-12

70.



Let R be the region bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{2}$, as shown in the figure above.

- Find the area of R .
- The region R is the base of a solid. For this solid, the cross sections perpendicular to the x -axis are squares. Find the volume of this solid.
- Write, but do not evaluate, an integral expression for the volume of the solid generated when R is rotated about the horizontal line $y = 2$.

Part A

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\text{Area} = \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right) dx = \frac{2}{3} x^{3/2} - \frac{x^2}{4} \Big|_{x=0}^{x=4} = \frac{4}{3}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\text{Area} = \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right) dx = \frac{2}{3} x^{3/2} - \frac{x^2}{4} \Big|_{x=0}^{x=4} = \frac{4}{3}$$

Part B

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

$$\begin{aligned} \text{Volume} &= \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right)^2 dx = \int_0^4 \left(x - x^{3/2} + \frac{x^2}{4} \right) dx \\ &= \frac{x^2}{2} - \frac{2x^{5/2}}{5} + \frac{x^3}{12} \Big|_{x=0}^{x=4} = \frac{8}{15} \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for antiderivative

1 point is earned for answer

8-11-12

$$\begin{aligned}\text{Volume} &= \int_0^4 \left(\sqrt{x} - \frac{x}{2} \right)^2 dx = \int_0^4 \left(x - x^{3/2} + \frac{x^2}{4} \right) dx \\ &= \frac{x^2}{2} - \frac{2x^{5/2}}{5} + \frac{x^3}{12} \bigg|_{x=0}^{x=4} = \frac{8}{15}\end{aligned}$$

Part C

1 point is earned for the limits and constant

2 points are earned for the integrand

$$\text{Volume} = \pi \int_0^4 \left(\left(2 - \frac{x}{2} \right)^2 - (2 - \sqrt{x})^2 \right) dx$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

2 points are earned for the integrand

$$\text{Volume} = \pi \int_0^4 \left(\left(2 - \frac{x}{2} \right)^2 - (2 - \sqrt{x})^2 \right) dx$$

8-11-12

71.  Let R be the region in the first quadrant bounded by the graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$.
- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is rotated about the vertical line $x = -1$.
- (c) The region R is the base of a solid. For this solid, the cross sections perpendicular to the y -axis are squares. Find the volume of this solid.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = 4.5$$

OR

$$\int_0^3 (3y - y^2) dy = 4.5$$

The graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$ intersect at the points $(0, 0)$ and $(9, 3)$.



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\int_0^9 \left(\sqrt{x} - \frac{x}{3} \right) dx = 4.5$$

OR

$$\int_0^3 (3y - y^2) dy = 4.5$$

The graphs of $y = \sqrt{x}$ and $y = \frac{x}{3}$ intersect at the points $(0, 0)$ and $(9, 3)$.

Part B

1 point is earned for constant and limits

2 points are earned for integrand

1 point is earned for the answer

$$\begin{aligned} \pi \int_0^3 \left((3y+1)^2 - (y^2+1)^2 \right) dy \\ = \frac{207\pi}{5} = 130.061 \text{ or } 130.062 \end{aligned}$$



0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for constant and limits

2 points are earned for integrand

1 point is earned for the answer

8-11-12

$$\pi \int_0^3 \left((3y+1)^2 - (y^2+1)^2 \right) dy$$

$$= \frac{207\pi}{5} = 130.061 \text{ or } 130.062$$

Part C

1 point is earned for integrand

1 point is earned for limits and answer

$$\int_0^3 (3y - y^2)^2 dy = 8.1$$



0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for integrand

1 point is earned for limits and answer

$$\int_0^3 (3y - y^2)^2 dy = 8.1$$

8-11-12

72. Let f be the function given by $f(x) = e^{-x}$, and let g be the function given by $g(x) = kx$, where k is the nonzero constant such that the graph of f is tangent to the graph of g .
- (a) Find the x -coordinate of the point of tangency and the value of k .
- (b) Let R be the region enclosed by the y -axis and the graphs of f and g . Using the results found in part (a), determine the area of R .
- (c) Set up, but do not integrate, an integral expression in terms of a single variable for the volume of the solid generated by revolving the region R , given in part (b), about the x -axis.

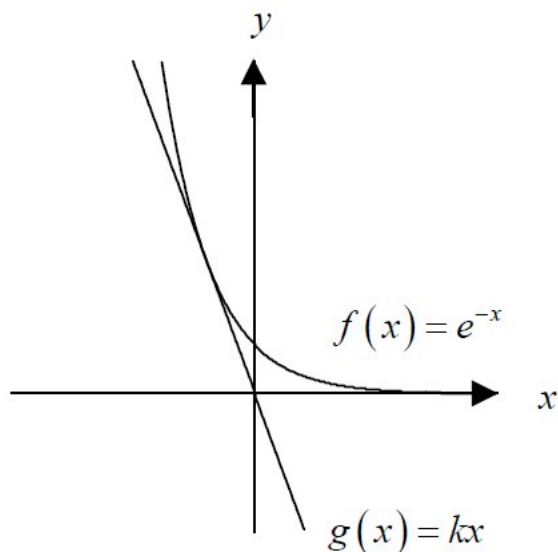
Part A

1 point is earned if finds $f'(x)$ and $g'(x)$ explicitly

1 point is earned for setting student's $f'(x)$ equal to student's $g'(x)$ explicitly

1 point is earned for setting $f(x) = g(x)$

1 point is earned for solving for x and k



$$f'(x) = -e^{-x}; g'(x) = k$$

$$-e^{-x} = k \leftarrow +2$$

$$e^{-x} = kx$$

$$x = -1 \text{ and } k = -e$$

✓

0	1	2	3	4
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The student response earns all of the following points:

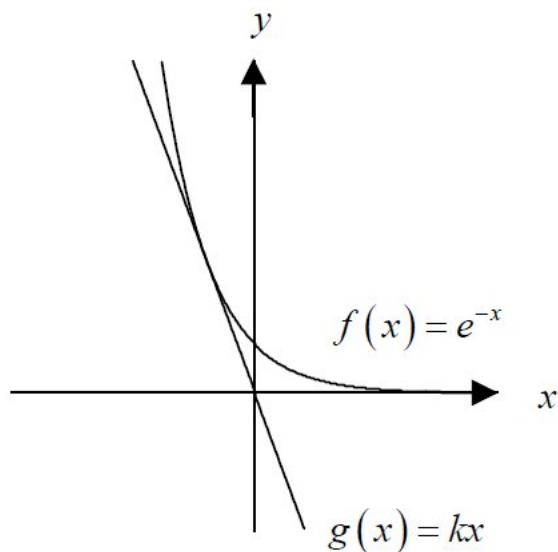
1 point is earned if finds $f'(x)$ and $g'(x)$ explicitly

1 point is earned for setting student's $f'(x)$ equal to student's $g'(x)$ explicitly

1 point is earned for setting $f(x) = g(x)$

1 point is earned for solving for x and k

8-11-12



$$f'(x) = -e^{-x}; g'(x) = k$$

$$-e^{-x} = k \leftarrow +2$$

$$e^{-x} = kx$$

$$x = -1 \text{ and } k = -e$$

Part B

2 points are earned for integral

1 point is earned if anti-differentiates

$$\int_{-1}^0 (e^{-x} - (-ex)) dx = \int_{-1}^0 (e^{-x} + ex) dx$$

$$= \left(-e^{-x} + \frac{ex^2}{2} \right) \Big|_{-1}^0$$

$$= (-1 + 0) - \left(-e + \frac{e}{2} \right)$$

$$= \frac{e}{2} - 1$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integral

1 point is earned if anti-differentiates

$$\int_{-1}^0 (e^{-x} - (-ex)) dx = \int_{-1}^0 (e^{-x} + ex) dx$$

$$= \left(-e^{-x} + \frac{ex^2}{2} \right) \Big|_{-1}^0$$

$$= (-1 + 0) - \left(-e + \frac{e}{2} \right)$$

$$= \frac{e}{2} - 1$$

Part C

2 points are earned for setting up integral with correct integrand

1 point **deducted** for incorrect limits/error involving π

8-11-12

$$\pi \int_{-1}^0 \left((e^{-x})^2 - (-ex)^2 \right) dx$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

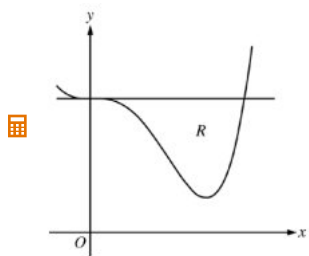
2 points are earned for setting up integral with correct integrand

1 point **deducted** for incorrect limits/error involving π

$$\pi \int_{-1}^0 \left((e^{-x})^2 - (-ex)^2 \right) dx$$

8-11-12

73.



Let R be the region enclosed by the graph of $f(x) = x^4 - 2.3x^3 + 4$ and the horizontal line $y = 4$, as shown in the figure above.

(a) Find the volume of the solid generated when R is rotated about the horizontal line $y = -2$.

(b) Region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is an isosceles right triangle with a leg in R . Find the volume of the solid.

(c) The vertical line $x = k$ divides R into two regions with equal areas. Write, but do not solve, an equation involving integral expressions whose solution gives the value k .

Part A

2 points are earned for the integrand

1 point is earned for limits

1 point is earned for the answer

$$f(x) = 4 \Rightarrow x = 0, 2.3$$

$$\begin{aligned} \text{Volume} &= \pi \int_0^{2.3} [(4+2)^2 - (f(x)+2)^2] dx \\ &= 98.868 \text{ (or } 98.867) \end{aligned}$$

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

1 point is earned for limits

1 point is earned for the answer

$$f(x) = 4 \Rightarrow x = 0, 2.3$$

$$\begin{aligned} \text{Volume} &= \pi \int_0^{2.3} [(4+2)^2 - (f(x)+2)^2] dx \\ &= 98.868 \text{ (or } 98.867) \end{aligned}$$

Part B

2 points are earned for integrand

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \int_0^{2.3} \frac{1}{2} (4 - f(x))^2 dx \\ &= 3.574 \text{ (or } 3.573) \end{aligned}$$

8-11-12

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^{2.3} \frac{1}{2}(4 - f(x))^2 dx \\ &= 3.574 \text{ (or } 3.573\text{)}\end{aligned}$$

Part C

1 point is earned for the area of one region

1 point is earned for the equation

$$\int_0^k (4 - f(x)) dx = \int_k^{2.3} (4 - f(x)) dx$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the area of one region

1 point is earned for the equation

$$\int_0^k (4 - f(x)) dx = \int_k^{2.3} (4 - f(x)) dx$$

8-11-12

74.  Let R be the region enclosed by the graph of $y = \sqrt{x-1}$, the vertical line $x = 10$, and the x -axis.
- (a) Find the area of R .
- (b) Find the volume of the solid generated when R is revolved about the horizontal line $y = 3$.
- (c) Find the volume of the solid generated when R is revolved about the vertical line $x = 10$.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_1^{10} \sqrt{x-1} \, dx = 18$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area} = \int_1^{10} \sqrt{x-1} \, dx = 18$$

Part B

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_1^{10} \left(9 - \left(3 - \sqrt{x-1} \right)^2 \right) dx$$

$$= 212.057 \text{ or } 212.058$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Volume} = \pi \int_1^{10} \left(9 - \left(3 - \sqrt{x-1} \right)^2 \right) dx$$

$$= 212.057 \text{ or } 212.058$$

Part C

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

8-11-12

$$\text{Volume} = \pi \int_0^3 (10 - (y^2 + 1))^2 dy$$

$$= 407.150$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

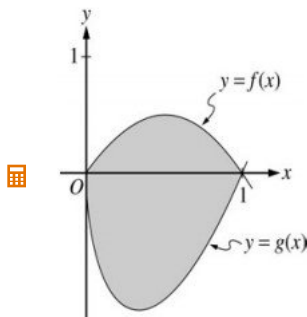
1 point is earned for the answer

$$\text{Volume} = \pi \int_0^3 (10 - (y^2 + 1))^2 dy$$

$$= 407.150$$

8-11-12

75.



Let f and g be the functions given by $f(x) = 2x(1 - x)$ and $g(x) = 3(x - 1)\sqrt{x}$ for $0 \leq x \leq 1$. The graphs of f and g are shown in the figure above.

(a) Find the area of the shaded region enclosed by the graphs of f and g .

(b) Find the volume of the solid generated when the shaded region enclosed by the graphs of f and g is revolved about the horizontal line $y = 2$.

(c) Let h be the function given by $h(x) = kx(1 - x)$ for $0 \leq x \leq 1$. For each $k > 0$, the region (not shown) enclosed by the graphs of h and g is the base of a solid with square cross sections perpendicular to the x -axis. There is a value of k for which the volume of this solid is equal to 15. Write, but do not solve, an equation involving an integral expression that could be used to find the value of k .

Part A

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^1 (f(x) - g(x)) dx \\ &= \int_0^1 (2x(1 - x) - 3(x - 1)\sqrt{x}) dx = 1.133 \end{aligned}$$

✓

0	1	2
---	---	---

The student response earns all of the following points:

1 point is earned for the integral

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^1 (f(x) - g(x)) dx \\ &= \int_0^1 (2x(1 - x) - 3(x - 1)\sqrt{x}) dx = 1.133 \end{aligned}$$

Part B

1 point is earned for the limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^1 \left((2 - g(x))^2 - (2 - f(x))^2 \right) dx \\ &= \pi \int_0^1 \left((2 - 3(x - 1)\sqrt{x})^2 - (2 - 2x(1 - x))^2 \right) dx \\ &= 16.179 \end{aligned}$$

8-11-12

✓

0	1	2	3	4
---	---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

2 points are earned for integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^1 \left((2 - g(x))^2 - (2 - f(x))^2 \right) dx \\ &= \pi \int_0^1 \left((2 - 3(x - 1)\sqrt{x})^2 - (2 - 2x(1 - x))^2 \right) dx \\ &= 16.179\end{aligned}$$

Part C

2 points are earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^1 (h(x) - g(x))^2 dx \\ \int_0^1 (kx(1 - x) - 3(x - 1)\sqrt{x})^2 dx &= 15\end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

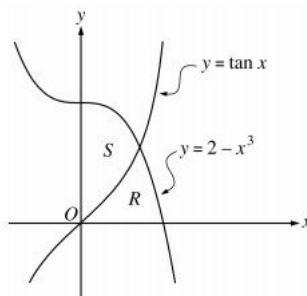
2 points are earned for the integrand

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^1 (h(x) - g(x))^2 dx \\ \int_0^1 (kx(1 - x) - 3(x - 1)\sqrt{x})^2 dx &= 15\end{aligned}$$

8-11-12

76.



Let R and S be the regions in the first quadrant shown in the figure above. The region R is bounded by the x -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$. The region S is bounded by the y -axis and the graphs of $y = 2 - x^3$ and $y = \tan x$.

- Find the area of R .
- Find the area of S .
- Find the volume of the solid generated when S is revolved about the x -axis.

Part A

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

Points of intersection

$$2 - x^3 = \tan x \text{ at } (A, B) = (0.902155, 1.265751)$$

$$\text{Area } R = \int_0^A \tan x \, dx + \int_A^{\sqrt[3]{2}} (2 - x^3) \, dx = 0.729$$

or

$$\text{Area } R = \int_0^B ((2 - y)^{1/3} - \tan^{-1} y) \, dy = 0.729$$

or

$$\text{Area } R = \int_0^{\sqrt[3]{2}} (2 - x^3) \, dx - \int_0^A (2 - x^3 - \tan x) \, dx = 0.729$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

Points of intersection

$$2 - x^3 = \tan x \text{ at } (A, B) = (0.902155, 1.265751)$$

$$\text{Area } R = \int_0^A \tan x \, dx + \int_A^{\sqrt[3]{2}} (2 - x^3) \, dx = 0.729$$

or

$$\text{Area } R = \int_0^B ((2 - y)^{1/3} - \tan^{-1} y) \, dy = 0.729$$

or

$$\text{Area } R = \int_0^{\sqrt[3]{2}} (2 - x^3) \, dx - \int_0^A (2 - x^3 - \tan x) \, dx = 0.729$$

8-11-12

Part B

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area } S = \int_0^A (2 - x^3 - \tan x) dx = 1.160 \text{ or } 1.161$$

or

$$\text{Area } S = \int_0^B \tan^{-1} y \, dy + \int_B^2 (2 - y)^{1/3} dy = 1.160 \text{ or } 1.161$$

or

$$\begin{aligned} \text{Area } S &= \int_0^2 (2 - y)^{1/3} dy - \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy \\ &= 1.160 \text{ or } 1.161 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits

1 point is earned for the integrand

1 point is earned for the answer

$$\text{Area } S = \int_0^A (2 - x^3 - \tan x) dx = 1.160 \text{ or } 1.161$$

or

$$\text{Area } S = \int_0^B \tan^{-1} y \, dy + \int_B^2 (2 - y)^{1/3} dy = 1.160 \text{ or } 1.161$$

or

$$\begin{aligned} \text{Area } S &= \int_0^2 (2 - y)^{1/3} dy - \int_0^B \left((2 - y)^{1/3} - \tan^{-1} y \right) dy \\ &= 1.160 \text{ or } 1.161 \end{aligned}$$

Part C

1 point is earned for the limits and constant

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Volume} &= \pi \int_0^A \left((2 - x^3)^2 - \tan^2 x \right) dx \\ &= 2.652\pi \text{ or } 8.331 \text{ or } 8.332 \end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the limits and constant

1 point is earned for the integrand

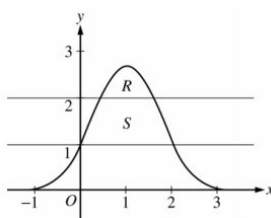
8-11-12

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left((2 - x^3)^2 - \tan^2 x \right) dx \\ &= 2.652\pi \text{ or } 8.331 \text{ or } 8.332\end{aligned}$$

8-11-12

77.



Let R be the region bounded by the graph of $y = e^{2x-x^2}$ and the horizontal line $y = 2$, and let S be the region bounded by the graph of $y = e^{2x-x^2}$ and the horizontal lines $y = 1$ and $y = 2$, as shown above.

- Find the area of R .
- Find the area of S .
- Write, but do not evaluate, an integral expression that gives the volume of the solid generated when R is rotated about the horizontal line $y = 1$.

Part A

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Area of } R = \int_P^Q (e^{2x-x^2} - 2)dx = 0.514$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Area of } R = \int_P^Q (e^{2x-x^2} - 2)dx = 0.514$$

Part B

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$e^{2x-x^2} = 1 \text{ when } x = 0, 2$$

8-11-12

$$\text{Area of } S = \int_0^2 (e^{2x-x^2} - 1)dx - \text{Area of } R$$

$$= 2.06016 - \text{Area of } R = 1.546$$

OR

$$\int_0^P (e^{2x-x^2} - 1)dx + (Q - P) \cdot 1 + \int_Q^2 (e^{2x-x^2} - 1)dx$$

$$= 0.219064 + 1.107886 + 0.219064 = 1.546$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the integrand

1 point is earned for the limits

1 point is earned for the answer

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

$$\text{Let } P = 0.446057 \text{ and } Q = 1.553943$$

$$e^{2x-x^2} = 1 \text{ when } x = 0, 2$$

$$\text{Area of } S = \int_0^2 (e^{2x-x^2} - 1)dx - \text{Area of } R$$

$$= 2.06016 - \text{Area of } R = 1.546$$

OR

$$\int_0^P (e^{2x-x^2} - 1)dx + (Q - P) \cdot 1 + \int_Q^2 (e^{2x-x^2} - 1)dx$$

$$= 0.219064 + 1.107886 + 0.219064 = 1.546$$

Part C

2 points are earned for the integrand

1 point is earned for the constant and limits

$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

$$\text{Let } P = 0.446057 \text{ and } Q = 1.553943$$

$$\text{Volume} = \pi \int_P^Q \left((e^{2x-x^2} - 1)^2 - (2 - 1)^2 \right) dx$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

8-11-12

2 points are earned for the integrand

1 point is earned for the constant and limits

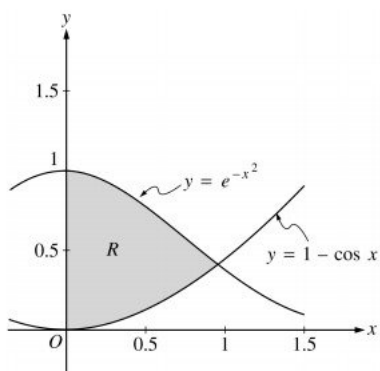
$$e^{2x-x^2} = 2 \text{ when } x = 0.446057, 1.553943$$

Let $P = 0.446057$ and $Q = 1.553943$

$$\text{Volume} = \pi \int_P^Q \left((e^{2x-x^2} - 1)^2 - (2 - 1)^2 \right) dx$$

8-11-12

78.



Let R be the shaded region in the first quadrant enclosed by the graphs of $y = e^{-x^2}$, $y = 1 - \cos x$, and the y -axis, as shown in the figure above.

- Find the area of the region R .
- Find the volume of the solid generated when the region R is revolved about the x -axis.
- The region R is the base of a solid. For this solid, each cross section perpendicular to the x -axis is a square. Find the volume of this solid.

Part A

1 point is earned for the correct limits in an integral in (a), (b), or (c).

Region R

$$e^{-x^2} = 1 - \cos x \text{ at } x = 0.941944 = A$$

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^A (e^{-x^2} - (1 - \cos x)) \, dx \\ &= 0.590 \text{ or } 0.591 \end{aligned}$$



0	1	2	3
---	---	---	---

The student response earns all of the following points:

1 point is earned for the correct limits in an integral in (a), (b), or (c).

Region R

$$e^{-x^2} = 1 - \cos x \text{ at } x = 0.941944 = A$$

1 point is earned for the integrand

1 point is earned for the answer

$$\begin{aligned} \text{Area} &= \int_0^A (e^{-x^2} - (1 - \cos x)) \, dx \\ &= 0.590 \text{ or } 0.591 \end{aligned}$$

Part B

2 points are earned for the integrand and constant

<-1> each error

1 point is earned for the answer

8-11-12

$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left((e^{-x^2})^2 - (1 - \cos x)^2 \right) dx \\ &= 0.55596\pi = 1.746 \text{ or } 1.747\end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand and constant

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \pi \int_0^A \left((e^{-x^2})^2 - (1 - \cos x)^2 \right) dx \\ &= 0.55596\pi = 1.746 \text{ or } 1.747\end{aligned}$$

Part C

2 points are earned for the integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^A \left(e^{-x^2} - (1 - \cos x) \right)^2 dx \\ &= 0.461\end{aligned}$$

✓

0	1	2	3
---	---	---	---

The student response earns all of the following points:

2 points are earned for the integrand

<-1> each error

1 point is earned for the answer

$$\begin{aligned}\text{Volume} &= \int_0^A \left(e^{-x^2} - (1 - \cos x) \right)^2 dx \\ &= 0.461\end{aligned}$$

8-11-12

79. Let M be the region in the first quadrant bounded above by the graph of $y = 3x$ and below by the graph of $y = x^2$. Which of the following gives the volume of the solid generated when region M is revolved about the vertical line $x = -3$?

(A) $\pi \int_0^9 \left((\sqrt{y} + 3) - \left(\frac{y}{3} + 3 \right) \right)^2 dy$

(B) $\pi \int_0^9 \left(\sqrt{y} - \frac{y}{3} \right) dy$

(C) $\pi \int_0^9 \left((\sqrt{y})^2 - \left(\frac{y}{3} \right)^2 \right) dy$

(D) $\pi \int_0^9 \left((\sqrt{y} + 3)^2 - \left(\frac{y}{3} + 3 \right)^2 \right) dy$ ✓

Answer D

Correct. Expressed as functions of y , the boundary curves are $x = \sqrt{y}$ and $x = \frac{y}{3}$ for $0 \leq y \leq 9$. Using the washer method, the outer radius of the solid is $R = \sqrt{y} - (-3) = \sqrt{y} + 3$ and the inner radius is $r = \frac{y}{3} - (-3) = \frac{y}{3} + 3$. Therefore the volume is given by the definite integral

$$\pi \int_0^9 (R^2 - r^2) dy = \pi \int_0^9 \left((\sqrt{y} + 3)^2 - \left(\frac{y}{3} + 3 \right)^2 \right) dy.$$

80. What is the volume of the solid generated when the circular region bounded by the graph of $x^2 + y^2 = 1$ is revolved around the horizontal line $y = 3$? (It is known that $\int_{-1}^1 \sqrt{1 - x^2} dx = \frac{\pi}{2}$.)

(A) π^2

(B) $\frac{16\pi}{3}$

(C) 12π

(D) $6\pi^2$ ✓

Answer D

Correct. The region is bounded above by the graph of $y = \sqrt{1 - x^2}$ and below by the graph of $y = -\sqrt{1 - x^2}$ for $-1 \leq x \leq 1$. Using the washer method, the outer radius is $R = 3 - (-\sqrt{1 - x^2}) = 3 + \sqrt{1 - x^2}$ and the inner radius is $r = 3 - \sqrt{1 - x^2}$. Therefore the volume is given by the definite integral

$$\pi \int_{-1}^1 (R^2 - r^2) dx = \pi \int_{-1}^1 \left((3 + \sqrt{1 - x^2})^2 - (3 - \sqrt{1 - x^2})^2 \right) dx, \text{ evaluated as follows.}$$

$$\pi \int_{-1}^1 \left((3 + \sqrt{1 - x^2})^2 - (3 - \sqrt{1 - x^2})^2 \right) dx = \pi \int_{-1}^1 12\sqrt{1 - x^2} dx = \pi \cdot 12 \cdot \frac{\pi}{2} = 6\pi^2$$

81.  Let R be the region in the first quadrant bounded by the graph of $y = 2 \tan\left(\frac{x}{5}\right)$, the line $y = 5 - x$, and the y -axis. What is the volume of the solid generated when R is revolved about the line $y = 6$?

(A) 27.345

(B) 66.723

(C) 92.103

(D) 209.617 ✓

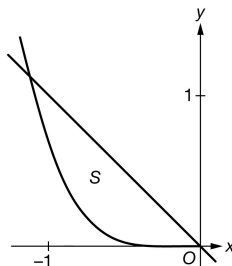
Answer D

Correct. It is helpful to use the graphing calculator to draw the two graphs. The calculator is also used to find the intersection point at $A = 3.389567$. Rotating a typical slice around the line $y = 6$ will form a washer with inner radius $r_1 = 6 - (5 - x) = 1 + x$ and outer radius $r_2 = 6 - 2 \tan\left(\frac{x}{5}\right)$. The calculator is used to numerically evaluate the integral expression for the volume.

$$\pi \int_0^A (r_2^2 - r_1^2) dx = \pi \int_0^A \left[\left(6 - 2 \tan\left(\frac{x}{5}\right) \right)^2 - (1 + x)^2 \right] dx = 209.617$$

8-11-12

82.



Let S be the region in the second quadrant bounded by the graphs of $y = x^2 \ln(1 - x^3)$ and $y = -x$, as shown in the figure above. The graphs intersect at $x = A$ and $x = 0$. Which of the following gives the volume of the solid generated when S is revolved about the line $y = -1$?

- (A) $\pi \int_A^0 ((-x + 1) - (x^2 \ln(1 - x^3) + 1))^2 dx$
- (B) $\int_A^0 ((-x + 1) - (x^2 \ln(1 - x^3) + 1))^2 dx$
- (C) $\pi \int_A^0 ((-x - 1)^2 - (x^2 \ln(1 - x^3) - 1)^2) dx$
- (D) $\pi \int_A^0 ((-x + 1)^2 - (x^2 \ln(1 - x^3) + 1)^2) dx$ ✓

Answer D

Correct. Revolving a typical slice about the line $y = -1$ will form a washer with inner radius $r_1 = x^2 \ln(1 - x^3) - (-1) = x^2 \ln(1 - x^3) + 1$ and outer radius $r_2 = -x - (-1) = -x + 1$. The integral expression for the volume is therefore

$$\begin{aligned} \int_A^0 (\pi r_2^2 - \pi r_1^2) dx \\ = \pi \int_A^0 ((-x + 1)^2 - (x^2 \ln(1 - x^3) + 1)^2) dx \end{aligned}$$

83. Let M be the region in the first quadrant bounded by the graph of $y = 3x + 2$, the graph of $y = x + 2$, and the vertical line $x = 3$. What is the volume of the solid generated when region M is revolved around the horizontal line $y = 1$?

- (A) 36π
- (B) 90π ✓
- (C) 108π
- (D) 126π

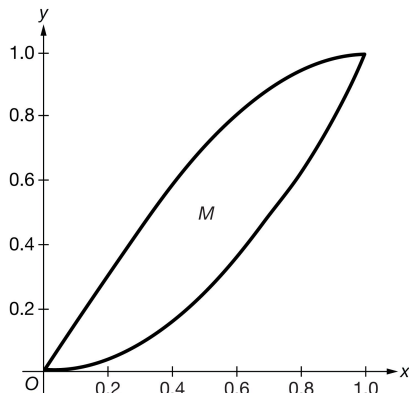
Answer B

Correct. Region M is bounded above by the graph of $y = 3x + 2$ and below by the graph of $y = x + 2$ for $0 \leq x \leq 3$. Using the washer method, the outer radius of the solid is $R = (3x + 2) - 1 = 3x + 1$ and the inner radius is $r = (x + 2) - 1 = x + 1$. Therefore the volume is given by the definite integral

$$\begin{aligned} \pi \int_0^3 (R^2 - r^2) dx &= \pi \int_0^3 ((3x + 1)^2 - (x + 1)^2) dx, \text{ evaluated as follows.} \\ \pi \int_0^3 ((3x + 1)^2 - (x + 1)^2) dx &= \pi \int_0^3 (8x^2 + 4x) dx = \pi \left(\frac{8}{3}x^3 + 2x^2 \right) \Big|_0^3 = \pi(72 + 18) = 90\pi \end{aligned}$$

8-11-12

84.



Let M be the region in the first quadrant bounded by the graphs of $y = \sin\left(\frac{\pi x}{2}\right)$ and $y = x^2$. What is the volume of the solid generated when region M is revolved around the vertical line $x = 2$?

- (A) 0.308
 (B) 0.953
 (C) 0.976
 (D) 2.836

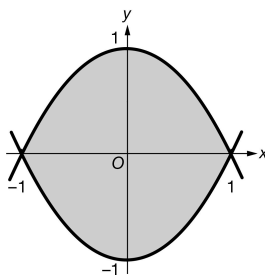
**Answer D**

Correct. Expressed as functions of y , the boundary curves are $x = \frac{2}{\pi} \sin^{-1} y$ and $x = \sqrt{y}$ for $0 \leq y \leq 1$. Using the washer method, the outer radius of the solid is $R = 2 - \frac{2}{\pi} \sin^{-1} y$ and the inner radius is $r = 2 - \sqrt{y}$. Therefore the volume is given by the definite integral

$$\pi \int_0^1 (R^2 - r^2) dy = \pi \int_0^1 \left(\left(2 - \frac{2}{\pi} \sin^{-1} y \right)^2 - (2 - \sqrt{y})^2 \right) dy.$$

This is evaluated using the graphing calculator.

85.



Let R be the region bounded above by the graph of $y = 1 - x^2$ and below by the graph of $y = x^2 - 1$, for $-1 \leq x \leq 1$, as shaded in the figure above. What is the volume of the solid generated when region R is revolved about the horizontal line $y = 3$?

- (A) $\frac{64\pi}{15}$
 (B) $\frac{8\pi}{3}$
 (C) 12π
 (D) 16π

**Answer D**

Correct. The region is bounded above by the graph of $y = 1 - x^2$ and below by the graph of $y = x^2 - 1$ for $-1 \leq x \leq 1$. Using the washer method, the outer radius is $R = (3 - (x^2 - 1)) = (4 - x^2)$ and the inner radius is $r = (3 - (1 - x^2)) = (2 + x^2)$.

Therefore, the volume is given by the definite integral $\pi \int_{-1}^1 (R^2 - r^2) dx = \pi \int_{-1}^1 ((4 - x^2)^2 - (2 + x^2)^2) dx$, which is evaluated as follows.

$$\pi \int_{-1}^1 ((4 - x^2)^2 - (2 + x^2)^2) dx = \pi \int_{-1}^1 ((16 - 8x^2 + x^4) - (4 + 4x^2 + x^4)) dx = \pi \int_{-1}^1 (12 - 12x^2) dx = \pi \left(12x - \frac{12x^3}{3} \right) \Big|_{-1}^1 = \pi((12 - 4) - (-12 + 4)) = 16\pi$$

8-11-12

86. Let M be the region in the first quadrant bounded below by the graph of $y = \frac{2}{\pi}x$ and above by the graph of $y = \sin x$. Which of the following gives the volume of the solid generated when region M is revolved about the vertical line $x = -3$?

(A) $\pi \int_0^1 \left(\left(\frac{\pi}{2}y + 3 \right) - (\arcsin y + 3) \right)^2 dy$

(B) $\pi \int_0^1 \left(\frac{\pi}{2}y - \arcsin y \right) dy$

(C) $\pi \int_0^1 \left(\left(\frac{\pi}{2}y \right)^2 - (\arcsin y)^2 \right) dy$

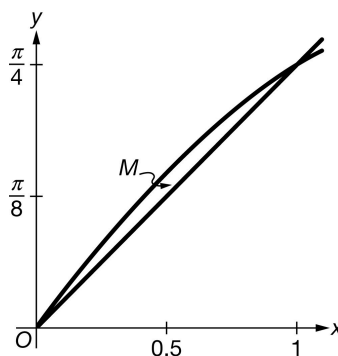
(D) $\pi \int_0^1 \left(\left(\frac{\pi}{2}y + 3 \right)^2 - (\arcsin y + 3)^2 \right) dy$ ✓

Answer D

Correct. Expressed as functions of y , the boundary curves are $x = \frac{\pi}{2}y$ and $x = \arcsin y$ for $0 \leq y \leq 1$. Using the washer method, the outer radius of the solid is $R = \frac{\pi}{2}y - (-3) = \frac{\pi}{2}y + 3$, and the inner radius is $r = \arcsin y - (-3) = \arcsin y + 3$. Therefore, the volume is given by the definite integral

$$\pi \int_0^1 (R^2 - r^2) dy = \pi \int_0^1 \left(\left(\frac{\pi}{2}y + 3 \right)^2 - (\arcsin y + 3)^2 \right) dy.$$

87.



- Let M be the region in the first quadrant bounded by the graphs of $y = \arctan x$ and $y = \frac{\pi}{4}x$, as shown in the figure above. What is the volume of the solid generated when region M is revolved around the vertical line $x = 2$?

(A) 0.010

(B) 0.145

(C) 0.148

(D) 0.431 ✓

Answer D

Correct. Expressed as functions of y , the boundary curves are $x = \tan y$ and $x = \frac{4}{\pi}y$ for $0 \leq y \leq \frac{\pi}{4}$. When the washer method is used, the outer radius of the solid is $R = 2 - \tan y$ and the inner radius is $r = 2 - \frac{4}{\pi}y$. Therefore, the volume is given by the definite integral

$$\pi \int_0^{\frac{\pi}{4}} (R^2 - r^2) dy = \pi \int_0^{\frac{\pi}{4}} \left((2 - \tan y)^2 - \left(2 - \frac{4}{\pi}y \right)^2 \right) dy. \text{ This is evaluated using the graphing calculator.}$$

8-11-12

88. Let M be the region in the first quadrant bounded by the graph of $y = 2x + 3$, the graph of $y = x + 3$, and the vertical line $x = 4$. What is the volume of the solid generated when region M is revolved about the horizontal line $y = 2$?

- (A) $\frac{64\pi}{3}$
 (B) 80π
 (C) 112π
 (D) 144π

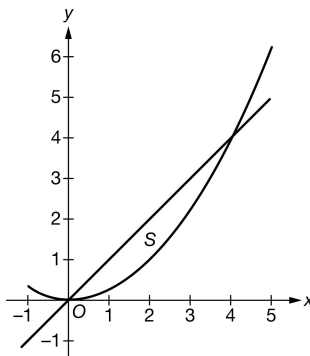
Answer B

Correct. Region M is bounded above by the graph of $y = 2x + 3$ and below by the graph of $y = x + 3$ for $0 \leq x \leq 4$. Using the washer method, the outer radius of the solid is $R = (2x + 3) - 2 = 2x + 1$ and the inner radius is $r = (x + 3) - 2 = x + 1$. Therefore, the volume is given by the definite integral

$$\pi \int_0^4 (R^2 - r^2) \, dx = \pi \int_0^4 ((2x + 1)^2 - (x + 1)^2) \, dx, \text{ evaluated as follows.}$$

$$\pi \int_0^4 ((2x + 1)^2 - (x + 1)^2) \, dx = \pi \int_0^4 (3x^2 + 2x) \, dx = \pi (x^3 + x^2) \Big|_0^4 = \pi (80 - 0) = 80\pi$$

89.



Let S be the region in the first quadrant bounded by the graphs of $y = \frac{x^2}{4}$ and $y = x$, as shown in the figure above. The graphs intersect at $x = 0$ and $x = 4$. Which of the following gives the volume of the solid generated when S is revolved about the line $y = -2$?

- (A) $\pi \int_0^4 \left((x + 2) - \left(\frac{x^2}{4} + 2 \right) \right)^2 \, dx$
 (B) $\int_0^4 \left((x + 2) - \left(\frac{x^2}{4} + 2 \right) \right)^2 \, dx$
 (C) $\pi \int_0^4 \left((x - 2)^2 - \left(\frac{x^2}{4} - 2 \right)^2 \right) \, dx$
 (D) $\pi \int_0^4 \left((x + 2)^2 - \left(\frac{x^2}{4} + 2 \right)^2 \right) \, dx$

Answer D

Correct. Revolving a typical slice about the line $y = -2$ will form a washer with inner radius $r_1 = \left(\frac{x^2}{4} \right) - (-2) = \frac{x^2}{4} + 2$ and outer radius $r_2 = x - (-2) = x + 2$. The integral expression for the volume is as follows.

$$\int_0^4 (\pi r_2^2 - \pi r_1^2) \, dx = \pi \int_0^4 \left((x + 2)^2 - \left(\frac{x^2}{4} + 2 \right)^2 \right) \, dx$$

8-11-12

90.  Let R be the region in the first quadrant bounded by the graph of $y = 2 \cos\left(\frac{x}{3}\right)$, the line $y = \frac{x}{4}$, and the y -axis. What is the volume of the solid generated when R is revolved about the line $y = 4$?
- (A) 12.535
(B) 18.438
(C) 23.636
(D) 74.255

Answer D

Correct. It is helpful to use the graphing calculator to draw the two graphs. The calculator is also used to find the intersection point at $A = 3.397$. Store this value in the calculator, and use the stored value in calculations. Rotating a typical slice around the line $y = 4$ will form a washer with inner radius $r_1 = 4 - 2 \cos\left(\frac{x}{3}\right)$ and outer radius $r_2 = 4 - \frac{x}{4}$. The calculator is used to numerically evaluate the integral expression for the volume.

$$\pi \int_0^A (r_2^2 - r_1^2) dx = \pi \int_0^A \left[\left(4 - \frac{x}{4}\right)^2 - \left(4 - 2 \cos\left(\frac{x}{3}\right)\right)^2 \right] dx = 74.255$$